ME-407

FINAL PRESENTATION

GROUP C-2

Design of Vertical Axis Wind Turbine

Group C2	
EGE SARP DENGIZMEN	BERK GAZELOGLU
MURAT BERK BUZLUK	PELIN DEMIREL
BERKE BAYRAKTAR	BERTUG MUMIN EKENEK
GOKBERK CICEK	OMER ONDER ASMASARI

Outline

Problem Definition

Design Requirements

Visualization

Original Aspects of the Design System

Operation of the Design System

Design Modifications

Engineering Analysis

Physical Components

Experience Gain

Future Work

PROBLEM DEFINITION

- CO_2 Emission & Climate Change
- High Levelized Cost of Electricity
- Lack of Renewable Energy Systems
- Making Wind Energy Accesible
- Unpredictible Wind Flow
- Generating Energy from Low Wind Speeds

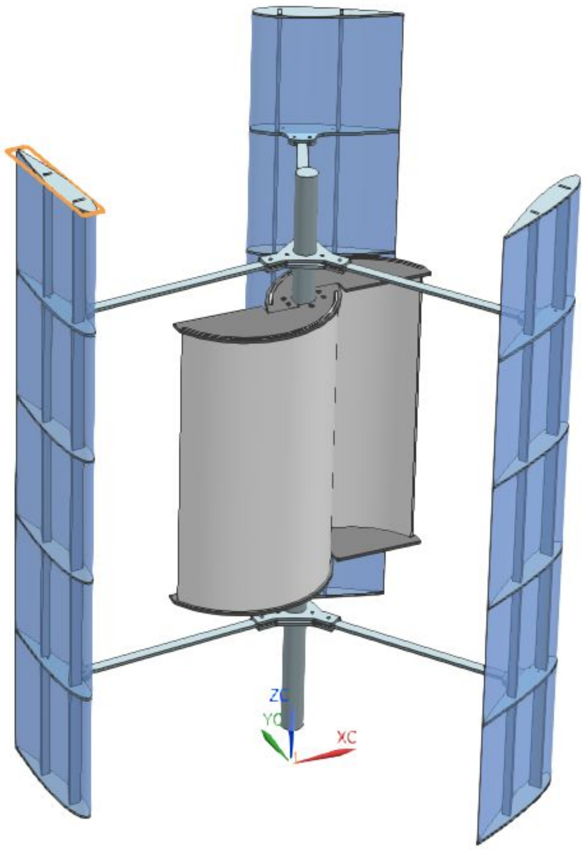
DESIGN REQUIREMENTS

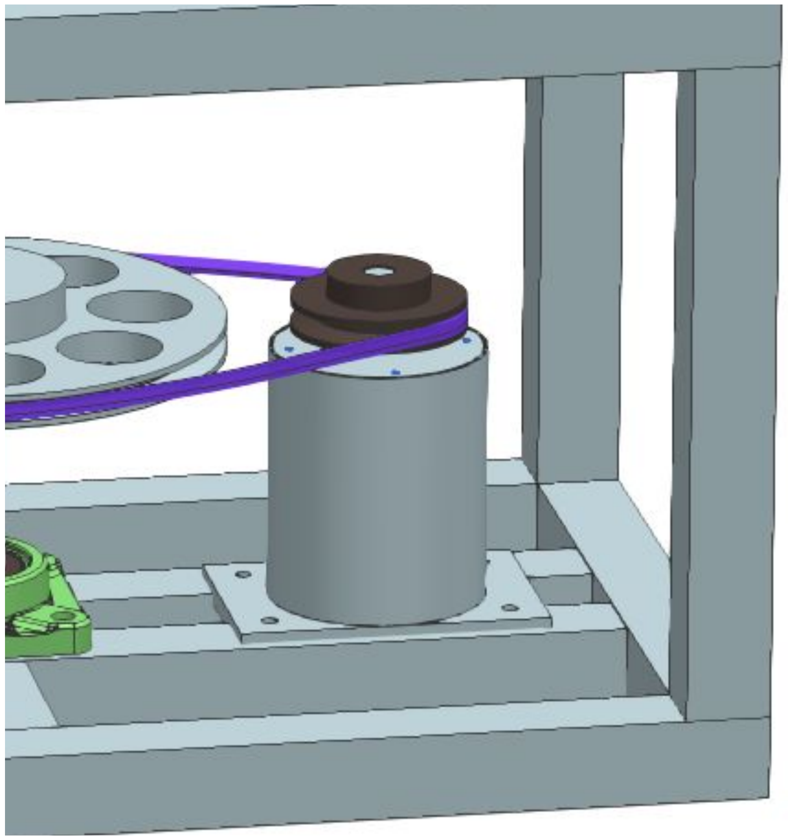
- Off grid
- Wind generation from various directions
- Appropiate for urban usage
- Easily manufacturable
- Easily maintained
- Meet the demand of small appliances
- «SHOULD BE TESTABLE AT THE AEROSPACE WIND TURBINE»

How Our Design Works

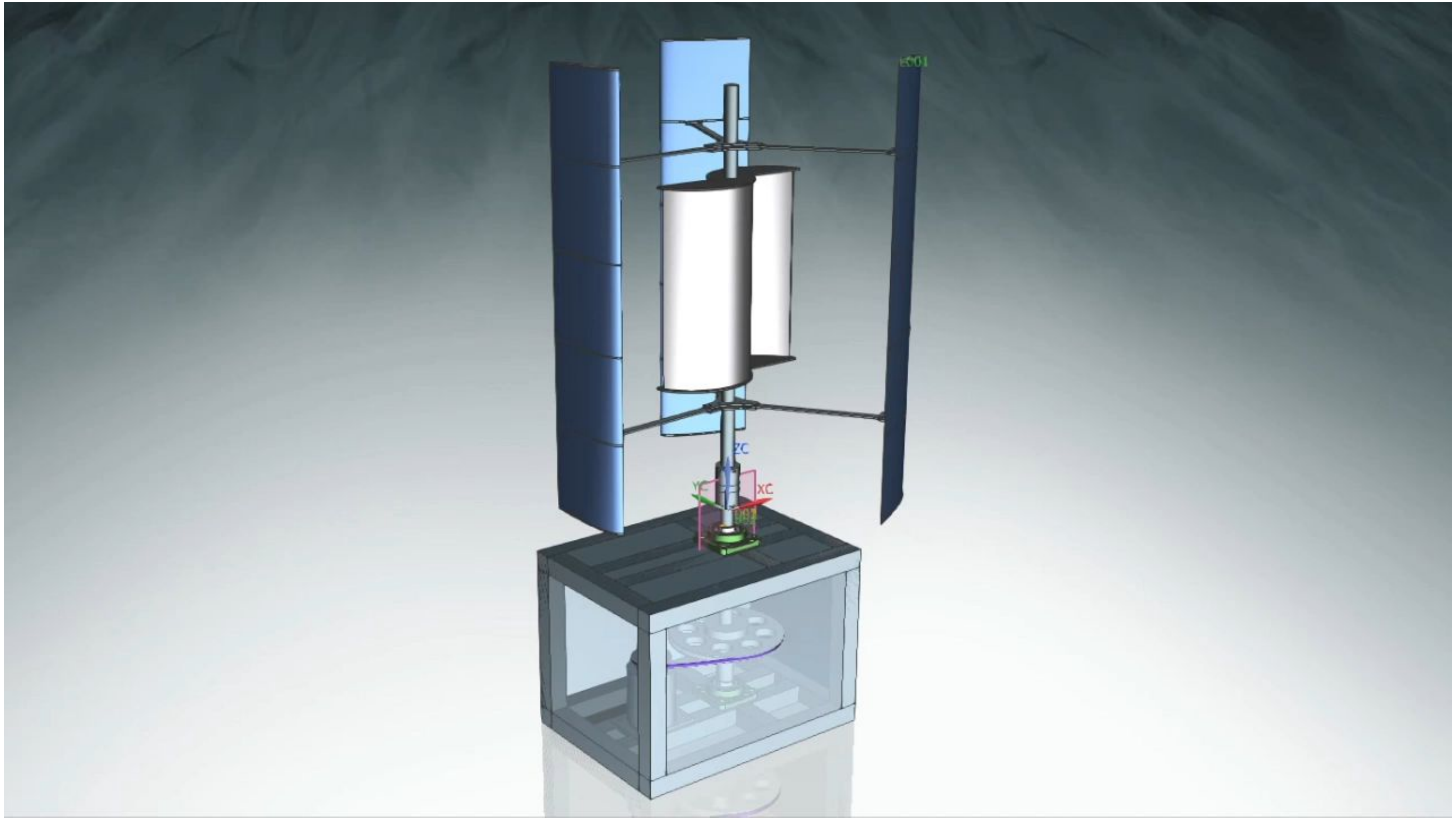
How Our Design Works

Capture Wind → Power Transmission → Generate Electricity

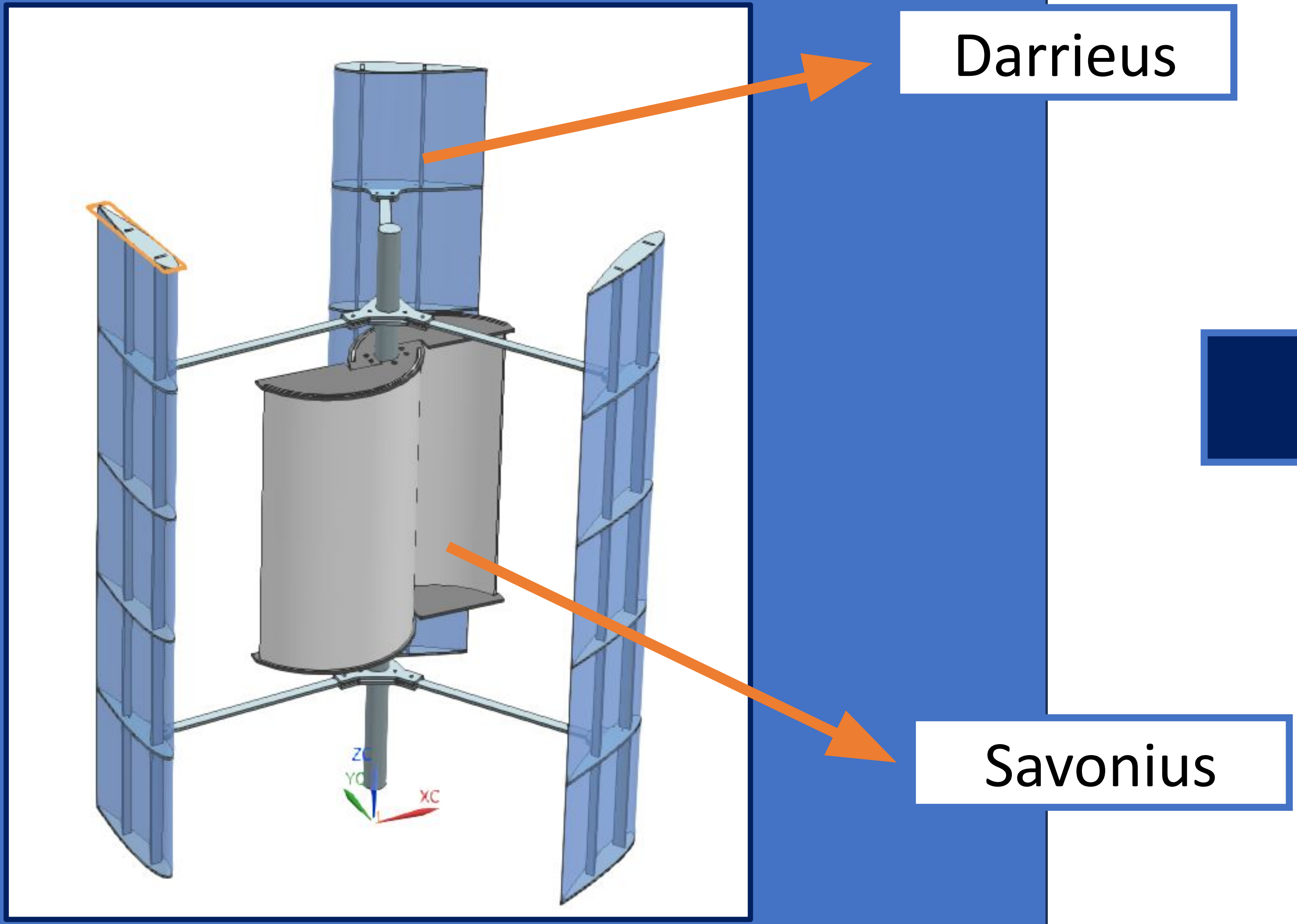




VISUALIZATION (All System)



How Our Design Works

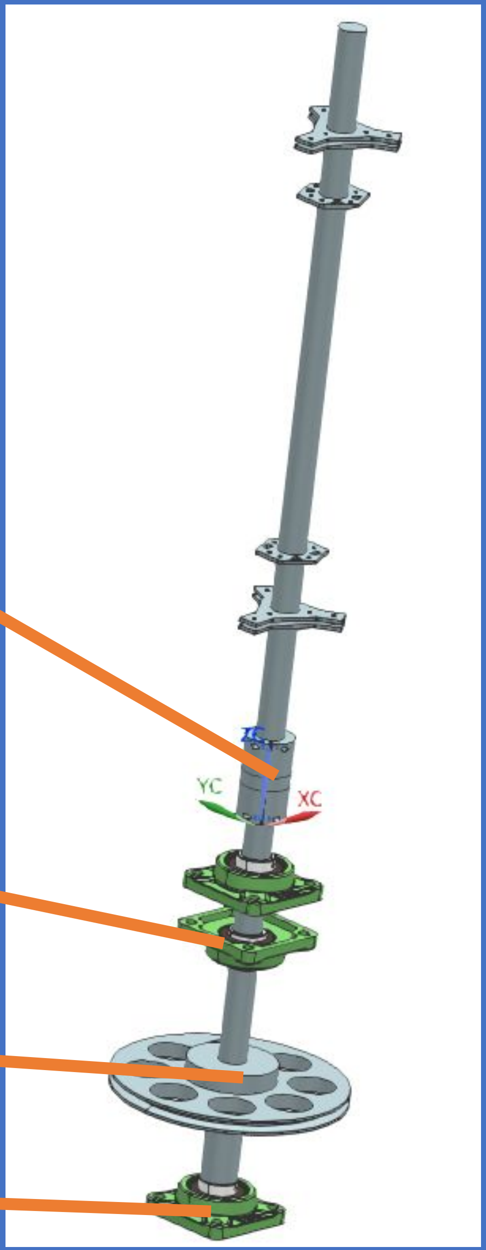


Capture Wind

How Our Design Works

Power Transmission

Coupling

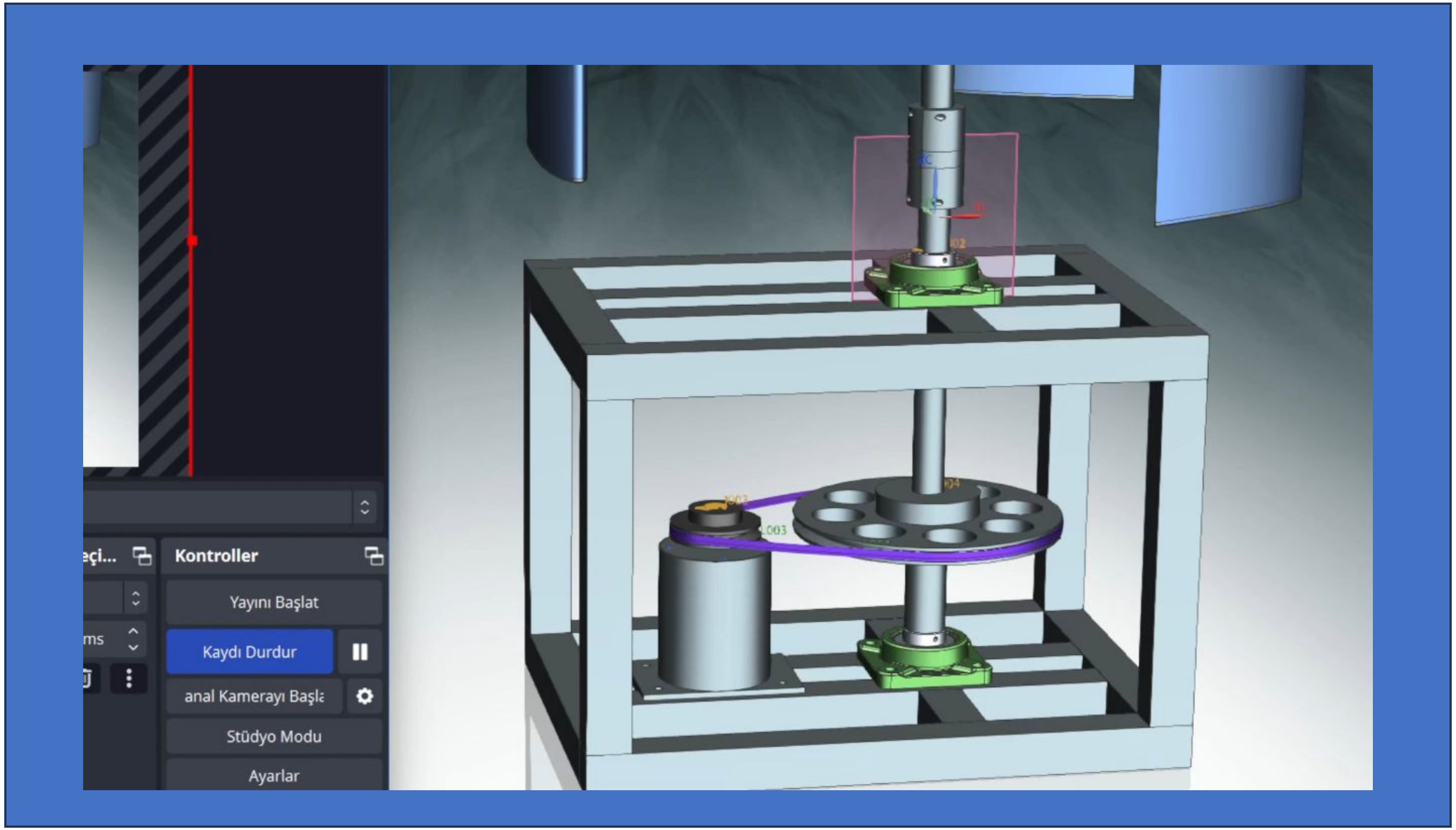


Bearing 1 and 2

V-Belt (SPZ 250)

Bearing 3

VISUALIZATION (V-Belt)



Performance Consideration

Design Criteria:

Cost	**25.85%**
Manufacturability	**20.85%**
Power Output	12.33%
Safety	10.25%
Low cut-in Speed	8.84%
Noise	8.69%
Durability	4.59%
Weight	4.47%
Size	4.33%

Cut-in Speed

V – Belt

Small Savonius

Power Output

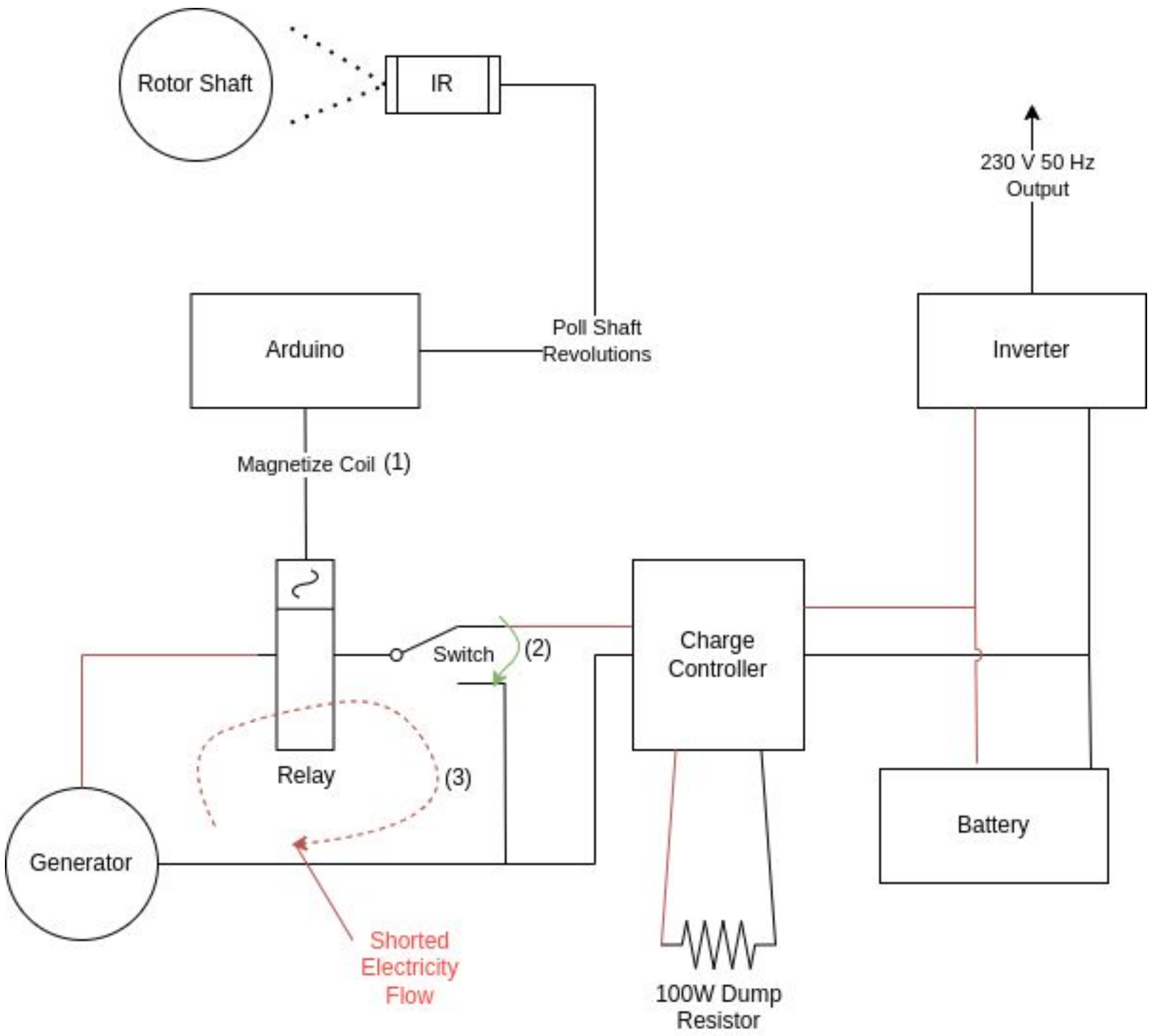
Generator Limits

Maximum Rotation

TSR and Solidity

TSR = 2 CP = 0.3

Electrical system

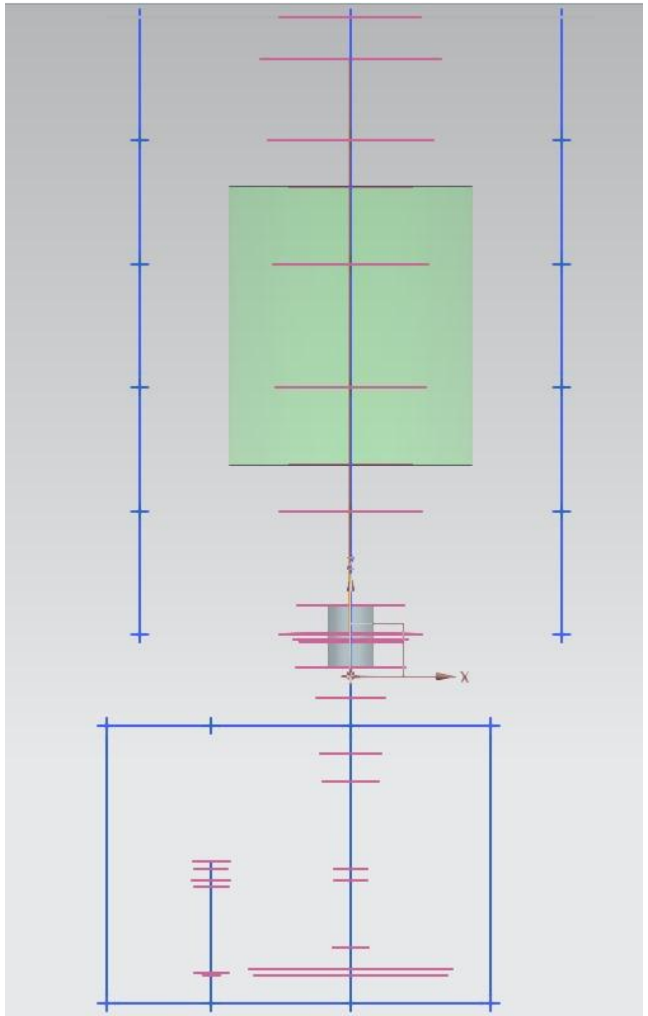


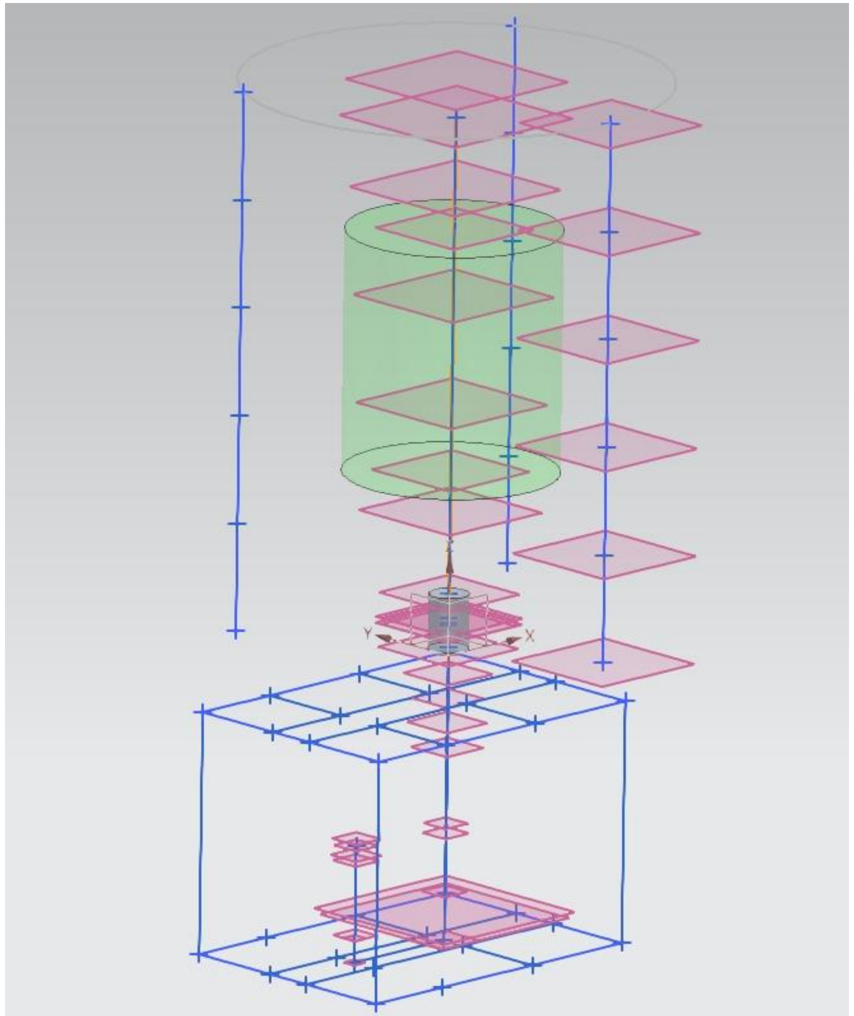
Components:

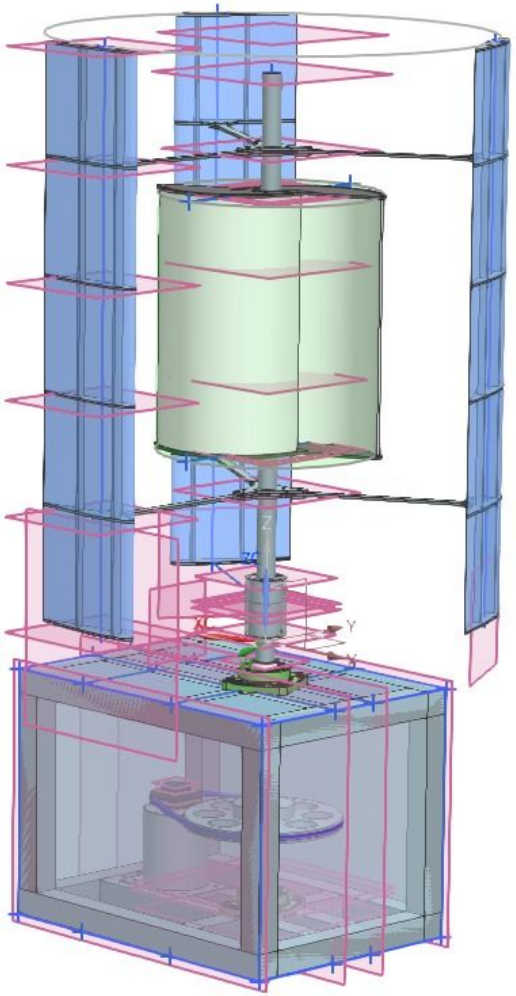
- MPPT Solar Charge Controller
- 12V/20Ah Battery
- 500W 12V Inverter
- 40V Brushed DC Motor
- 100W Dump Load Resistor
- Brake System Components (Explained in later sections)

ORIGINAL ASPECTS OF THE DESIGN SYSTEM

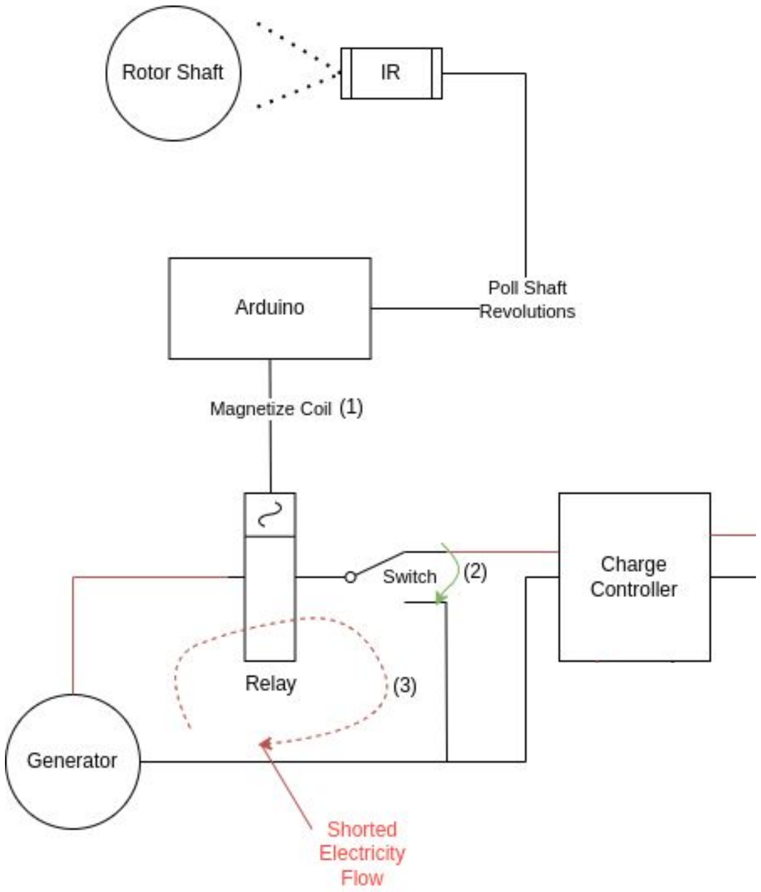
Layout Study







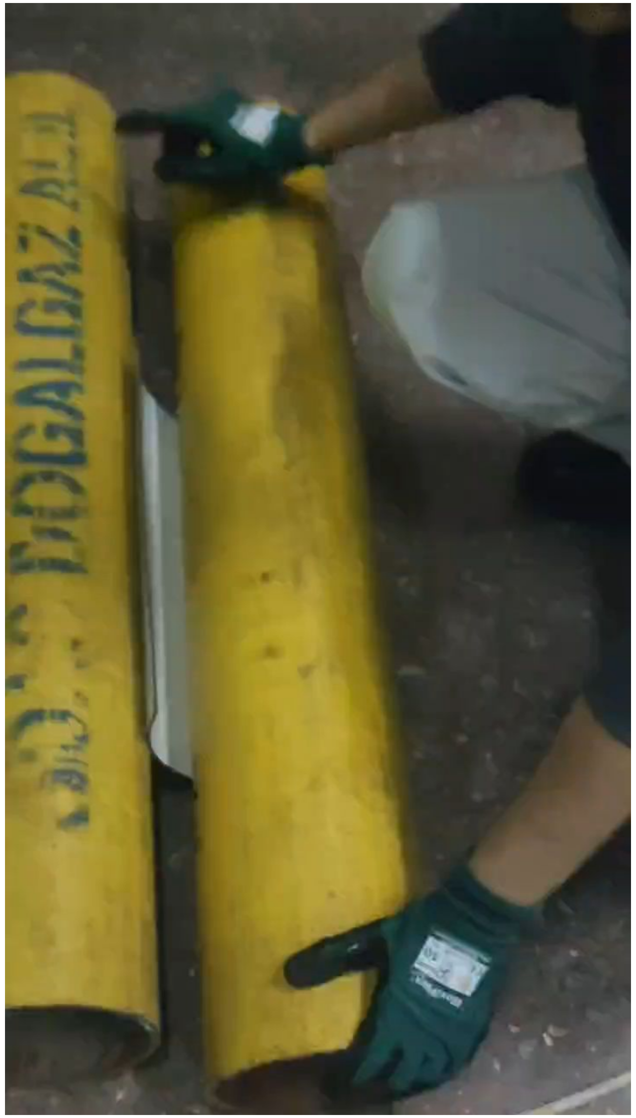
Brake System



Demonstration Video:

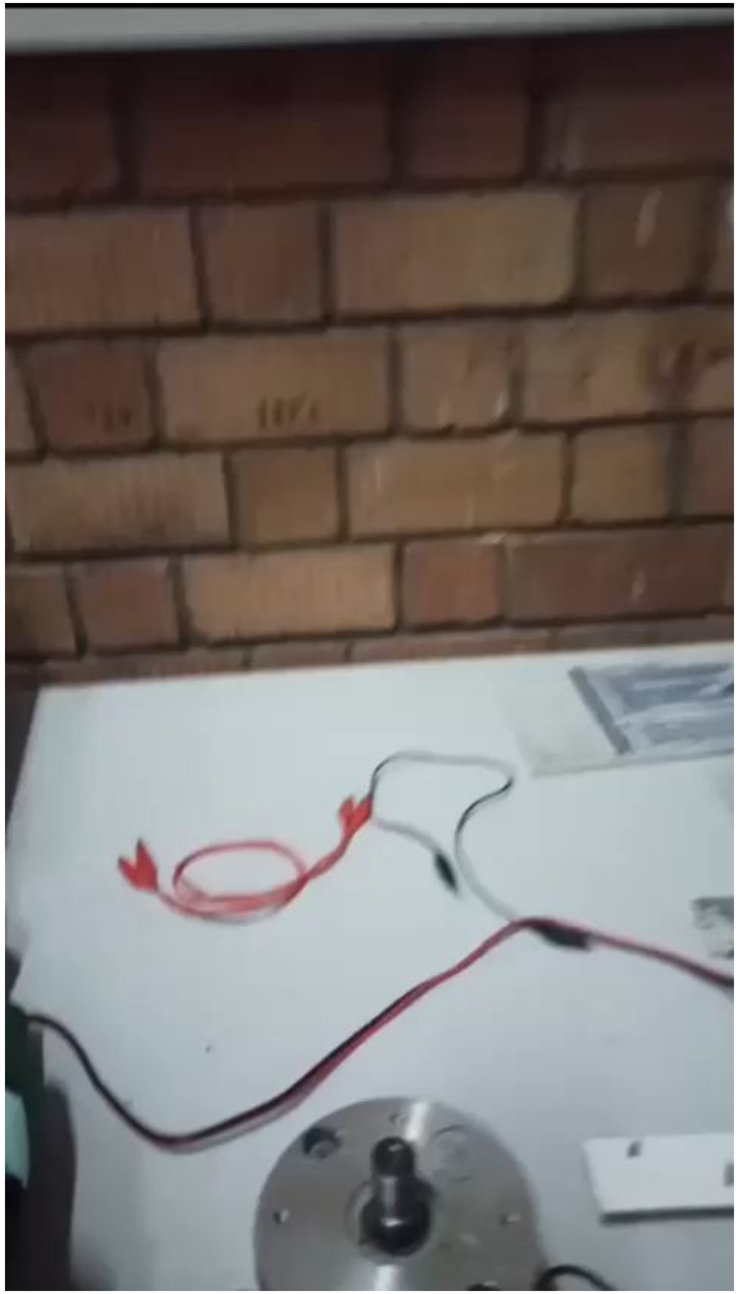
Aluminum Composite Bending

- A cheap and reliable material
- Can also be bended manually using steel pipes



Wire Cutting Machine

- Hard times creates strong people
- Strong people cut their foam own their own



Design Modifications

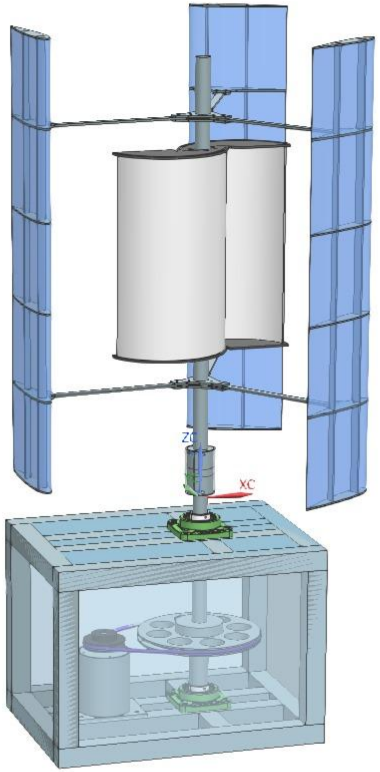
DESIGN MODIFICATIONS

Mock-Up Stage

Detailed Design

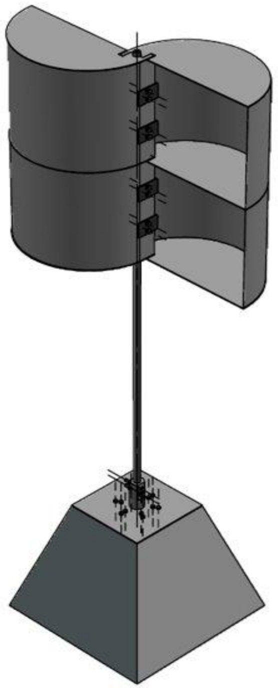
Production stage

Intermediate Stage



DESIGN MODIFICATIONS

Mock-Up Stage → **Detailed Design**



From Mock-Up to Detailed Design

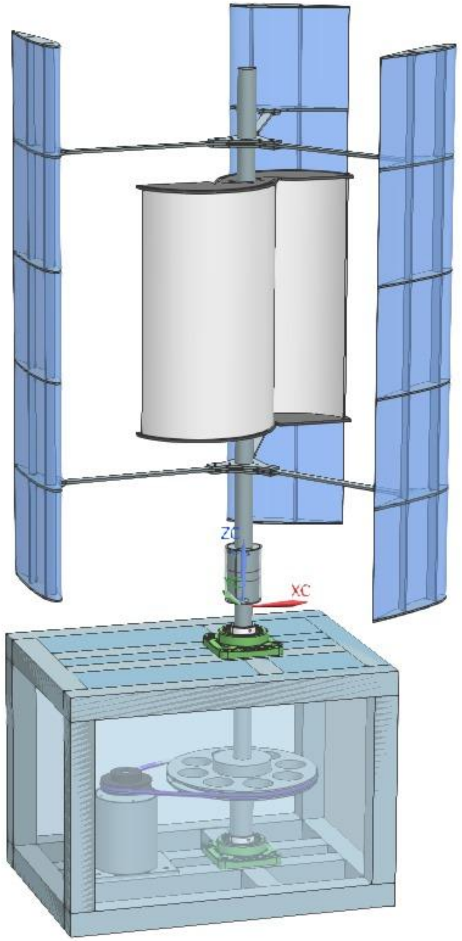
- **Blade type:** Two Savonius Blade to Hybrid Type Blade
  - Increase in power coefficient
- **Base Structure:** Pyramid Base to Square Base
  - Reinforced with square profile
- **Mechanical Energy Transmission:** Direct Drive to V-Belt
  - To achieve high rpm for generator
- **Additional Changes:** Bearing Addition
  - To decrease the vibration of shaft

DESIGN MODIFICATIONS

Detailed Design → **Production**

From Detailed Design to Production:

- **Base Structural Beams:** Square Beams to Aliminum Sigma Profile
  - Availability of Aliminum Profile & Ease of Assemble/Disassemble
- **Base Cover Plates:** Heavy Sheet Metal to Plexy Glass
  - Visible from outside
- **Savonius Blade:** FDM to Plexy Glass & Composite Material
  - Low cost & Out of charge Plexy and Composite
- **Blade Spar:** MDF to Composite
  - Higher structural integrity & Chordwise bending & Still light
- **Shaft Base Connection Rod:** Connection Rod to Double Bearing with Housing
  - Decrease in bearing radial load & reducing cost



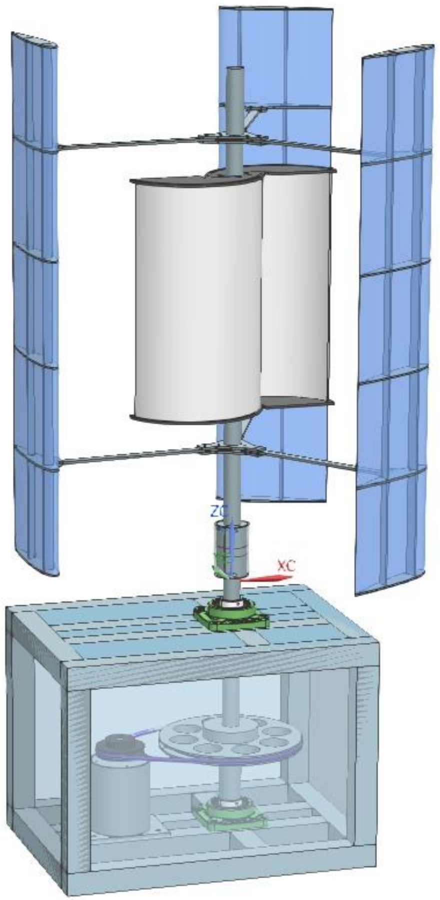
DESIGN MODIFICATIONS

Detailed Design → **Production**

From Detailed Design to Production:

- **Generator Holder:** Initial Load for V-Belt
    - Low cost

- **Coupling:** Can Be Tested Into Wind Tunnel
    - Modular Design

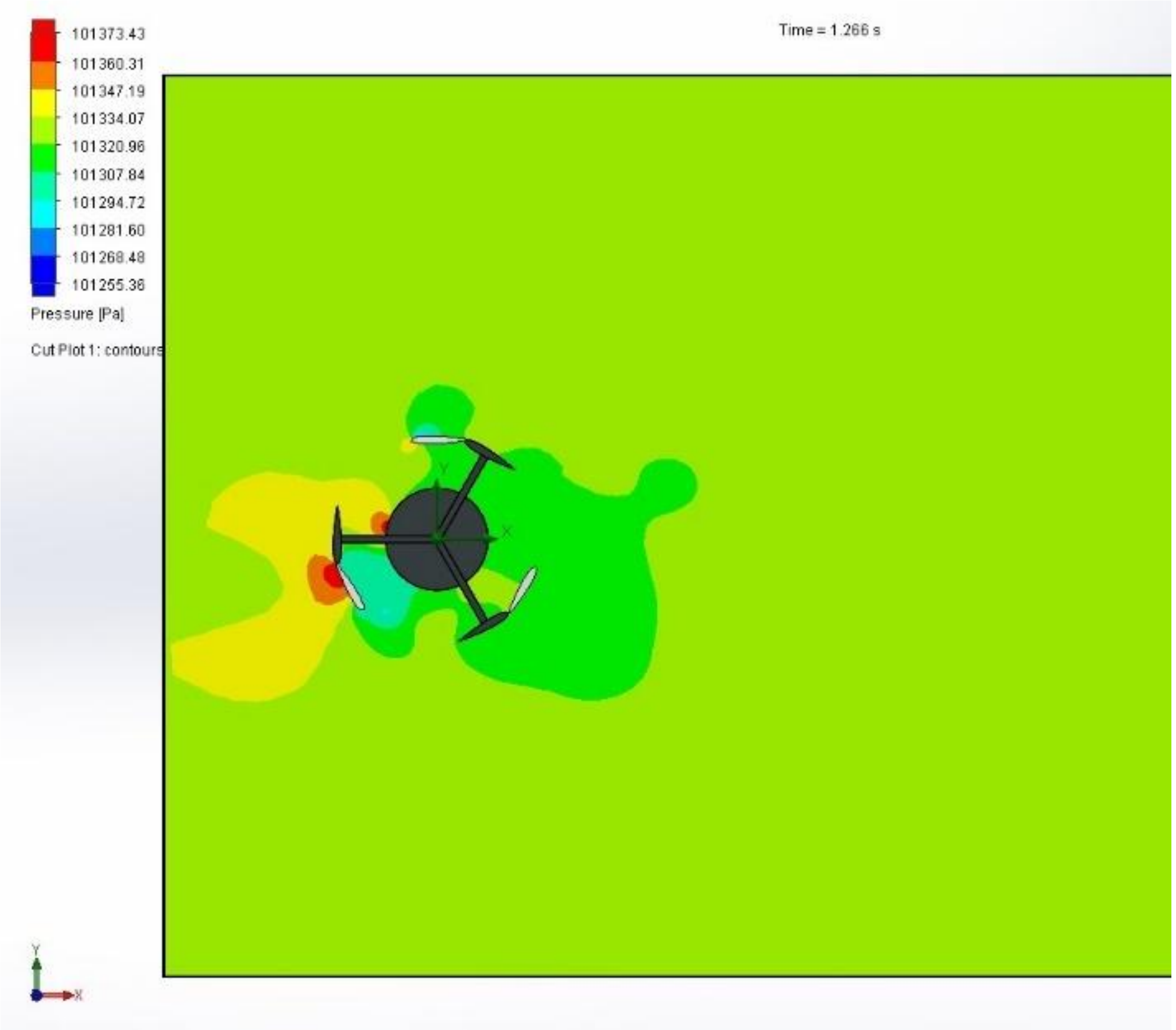


ENGINEERING ANALYSIS

CFD: Dimensioning

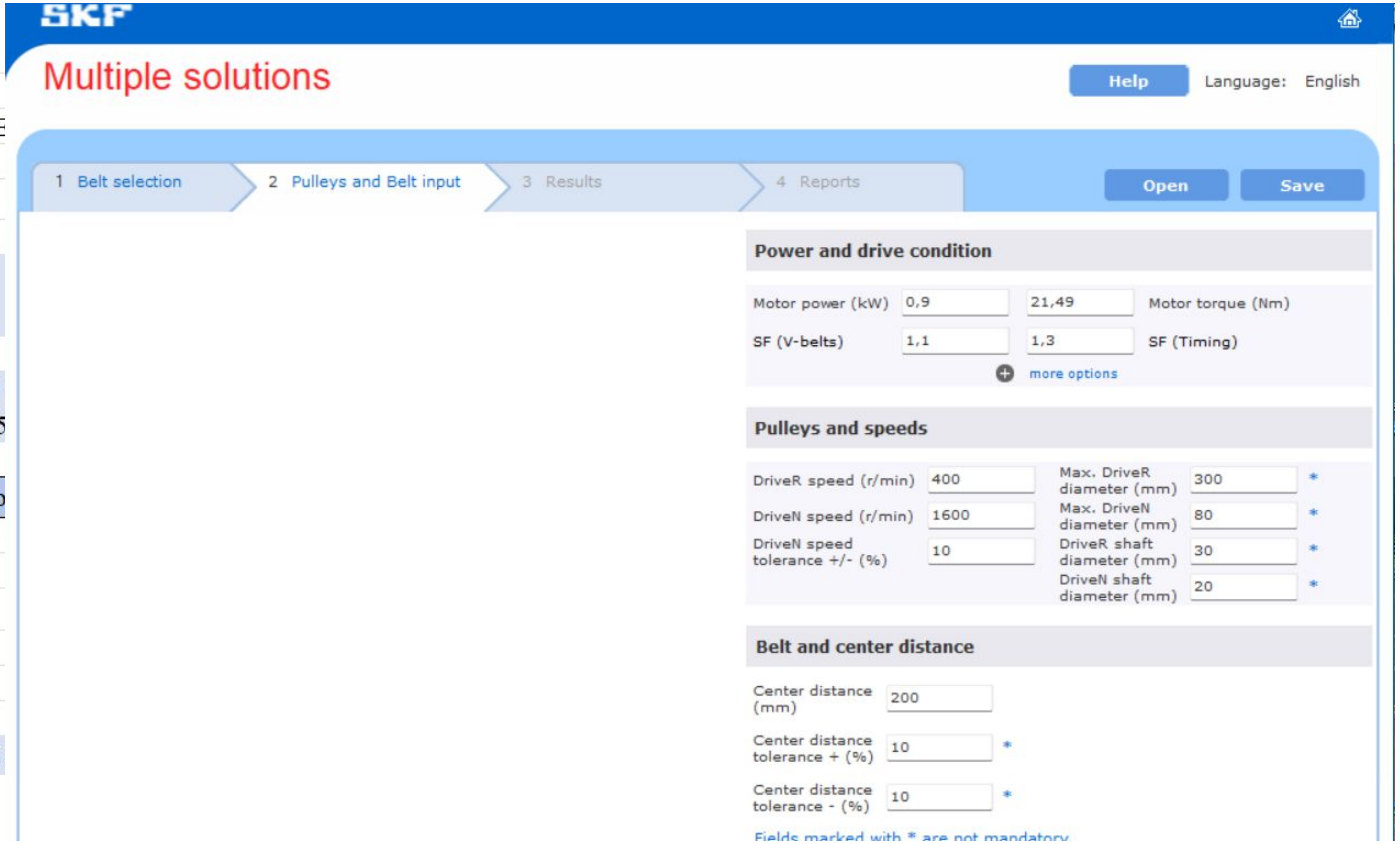
CFD: Inflicted Loads

- **Pressure Contour of Wind Turbine**

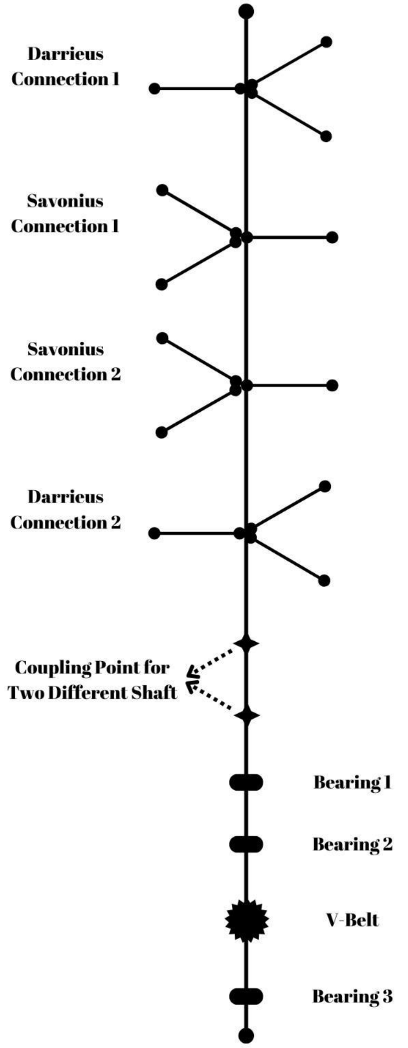


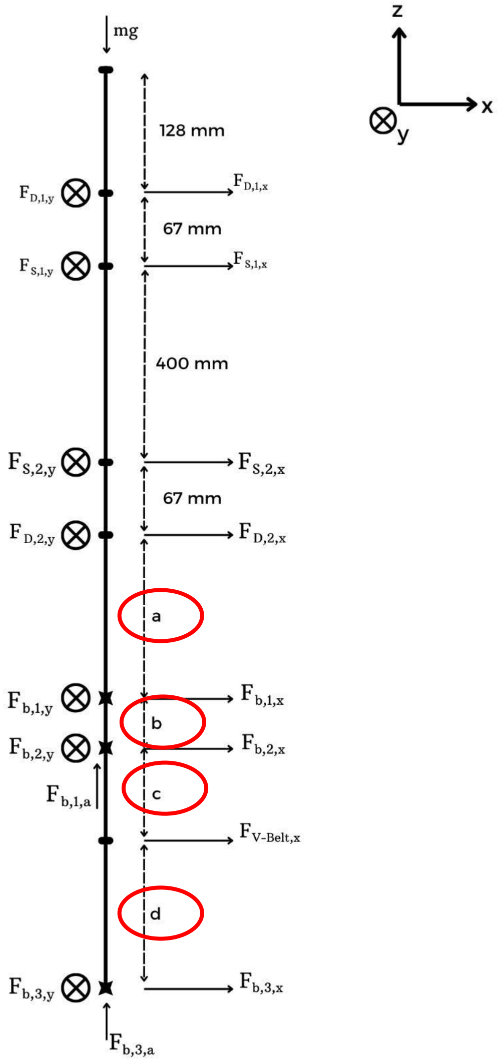
Selection of V-Belt and Pulleys

V-BELT SELECTION				
Oveload Service Factor	c2	1.1		(TABLE
Actual Power	P_B	0.99		
diameter of smaller sheave	d_{wk}	63	mm	
diameter of larger sheave	d_{wb}	252	mm	
Recommended Center Distance	e	max	630	
		min	220.5	
Effective Length (L_W)	Approximate	1239.356 mm		fro
	Exact	1239.753 mm		
f_B (frequency of strap)		9.491883 s^{-1}		
c3 from table 7		0.912126		
P_N Nominal Power		(from table 6)	1.33	



Bearing and Belt Location Optimization

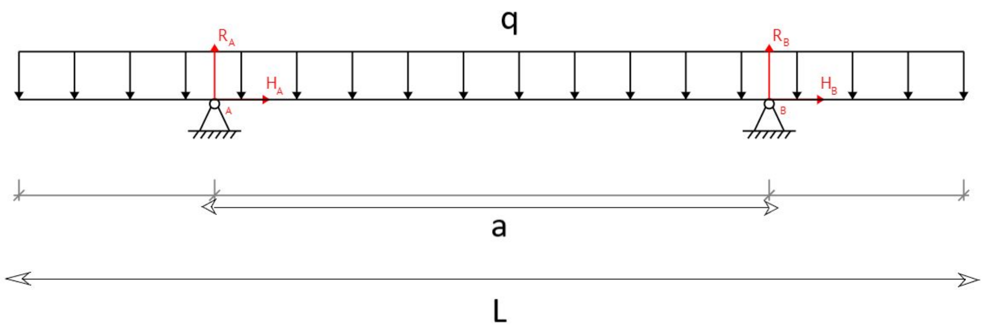




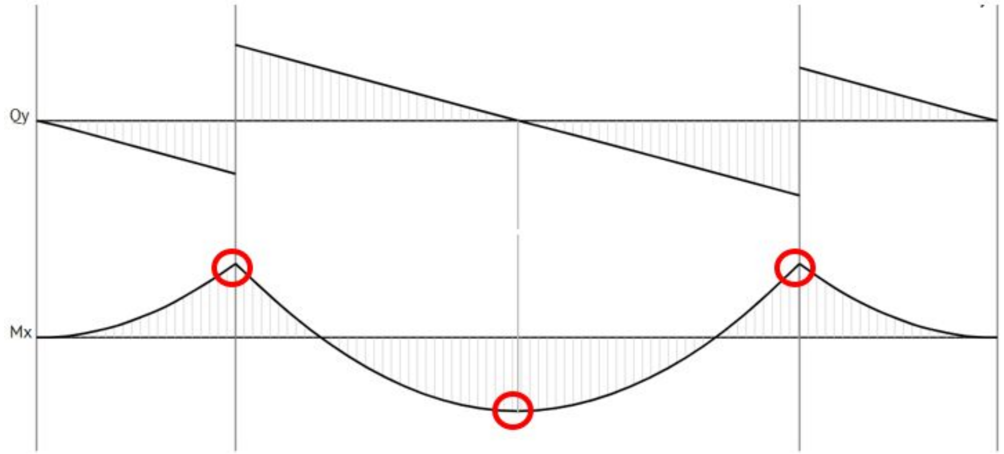
						F_{D1x}	316	N
						F_{S1x}	-90.7	N
	a	b	c	d		F_{S2x}	-90.7	N
678	268	60	140	210	350	F_{D2x}	316	N
min	268	30	100	100	400	F_{Vbelt}	-300	N
max	338	60	300	300				
						F_{D1y}	-3.11	N
$F_{x,total,bearing}$	-150.6	N				F_{S1y}	-4.299	N
$F_{y,total,bearing}$	14.818	N				F_{S2y}	-4.299	N
						F_{D2y}	-3.11	N

Optimize Darrieus Wing Connection

Free Body Diagram



Shear And Moment Diagrams

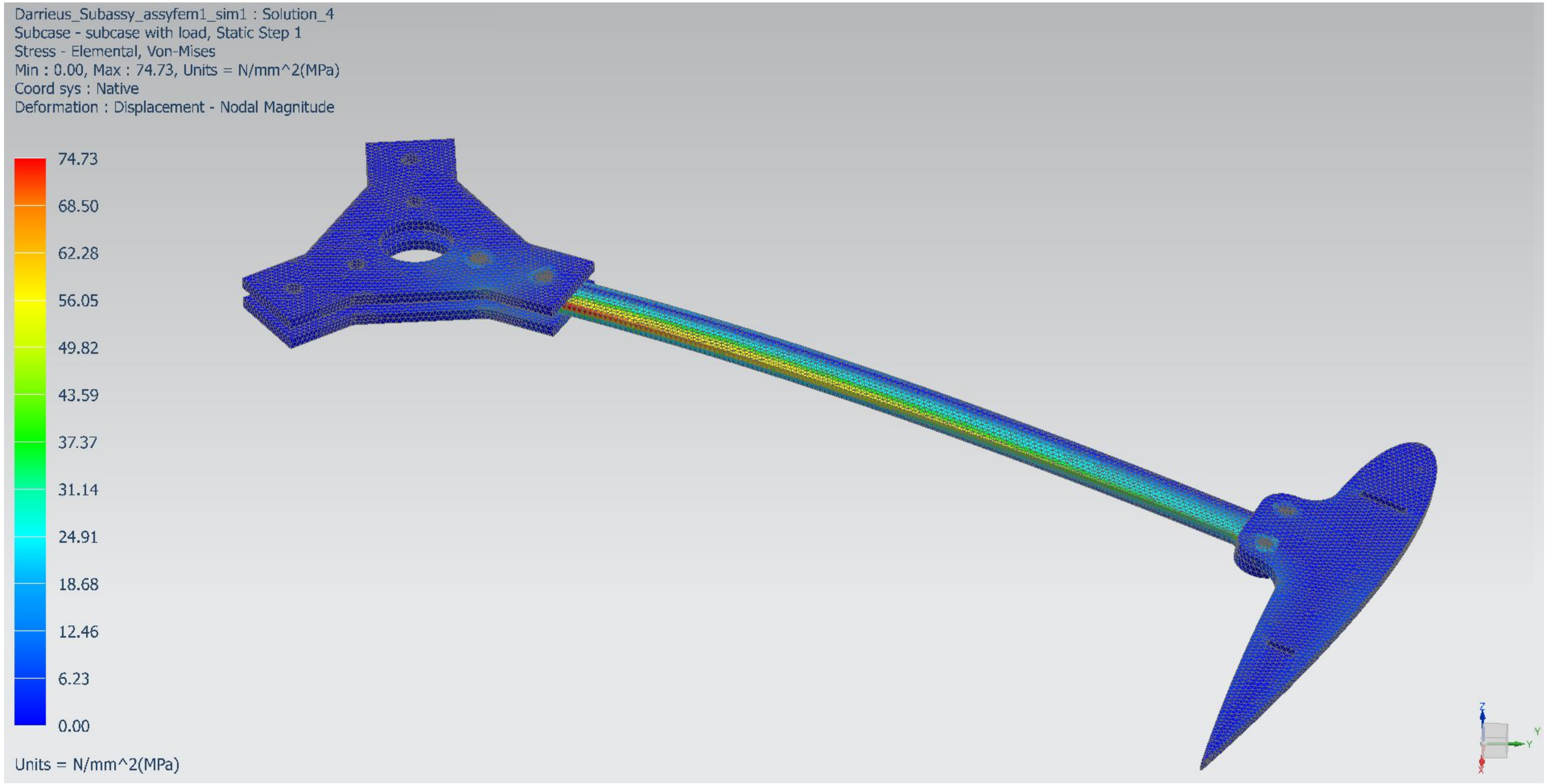


a/L ratio of 0.59 for min bending moment

Nastran Structural Analysis

Analysis of the Darrieus Wing Connection Pieces

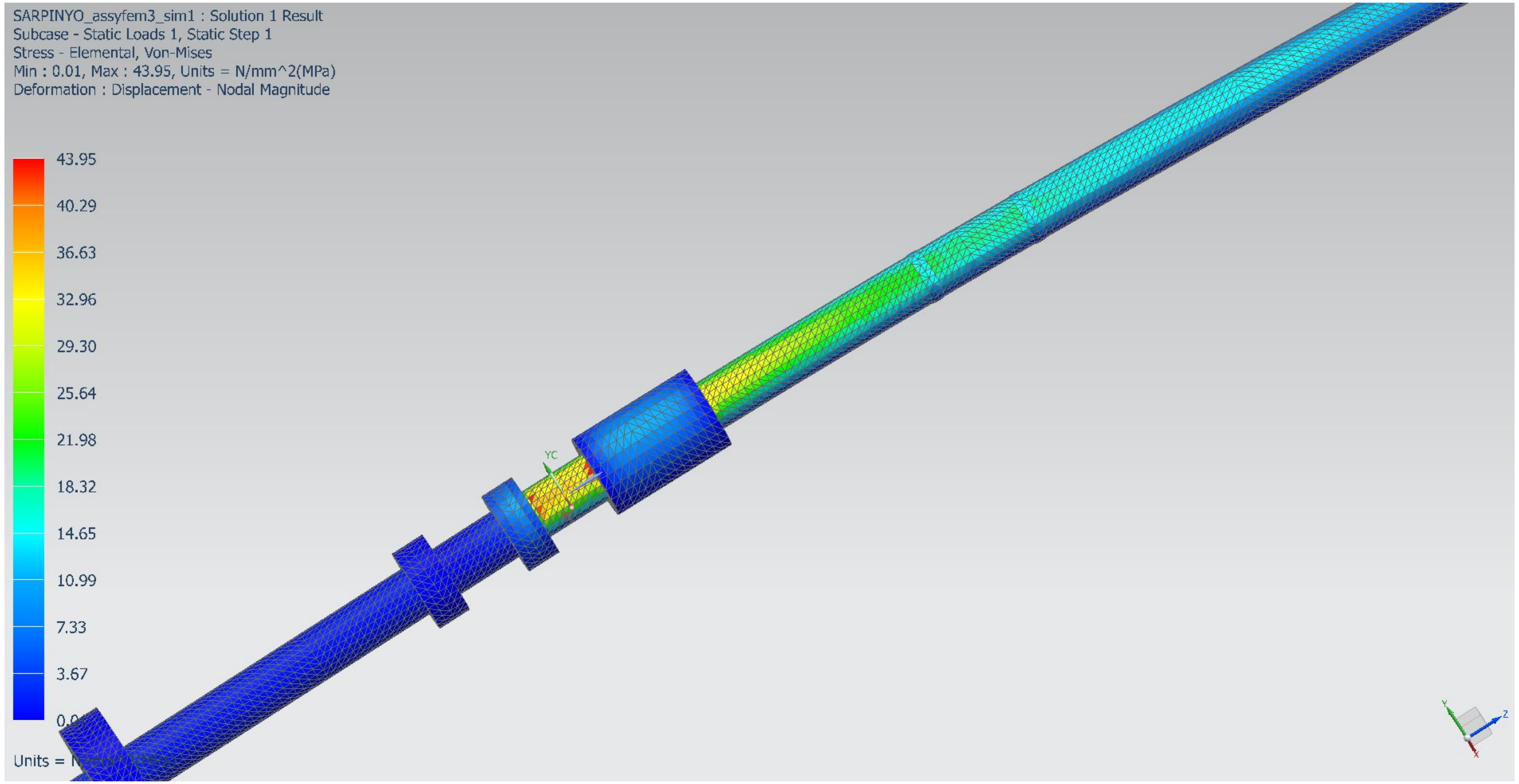
30 m/s wind speed static loading
400 rpm rotational speed
10 m/s angular deceleration



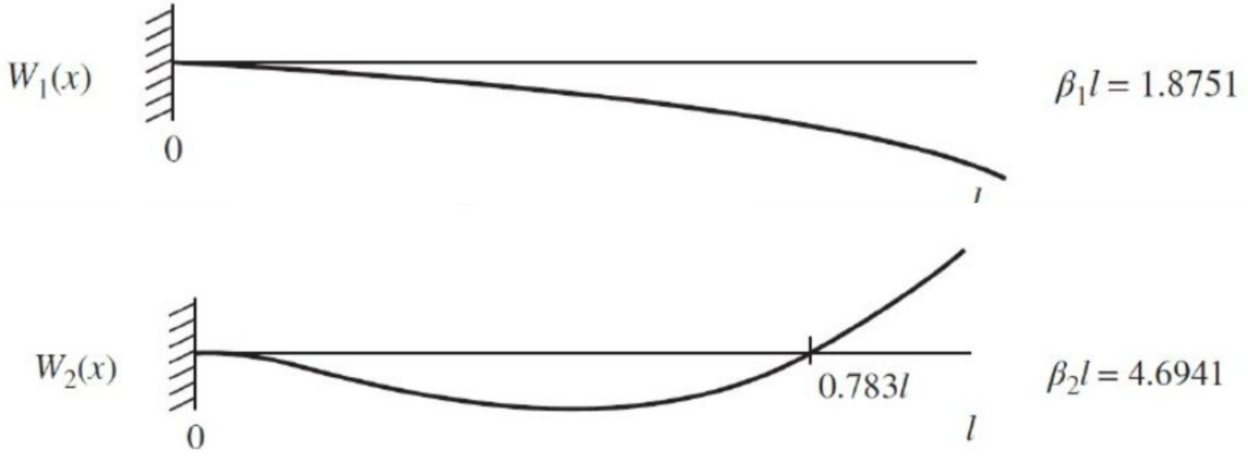
Nastran Structural Analysis

Analysis of the Shaft

30 m/s wind speed static loading
400 rpm rotational speed
10 m/s angular deceleration



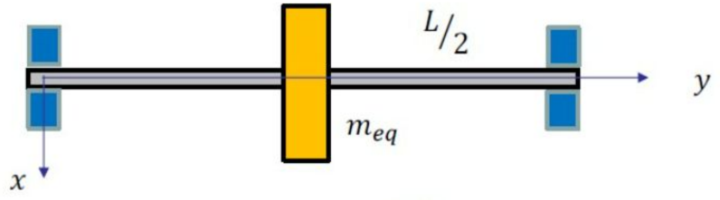
Vibration Analysis w/ Fixed End Continious Beam Assumption



$$EI\frac{\partial^4 w}{\partial x^4} + \rho A\frac{\partial^2 w}{\partial t^2} = f(x,t),$$

$$k_{eq} = \frac{3EI}{L^3}, \qquad m_{eq} = 0.24m$$

$$\omega_n = (\beta_n l)^2\sqrt{\frac{EI}{\rho A l^4}} \quad where\; n = 1,2,3,\ldots,$$



$$First\ Mode\ Natural\ Frequency \approx 132\frac{rad}{s} \approx 1263\ rpm$$

Physical Components

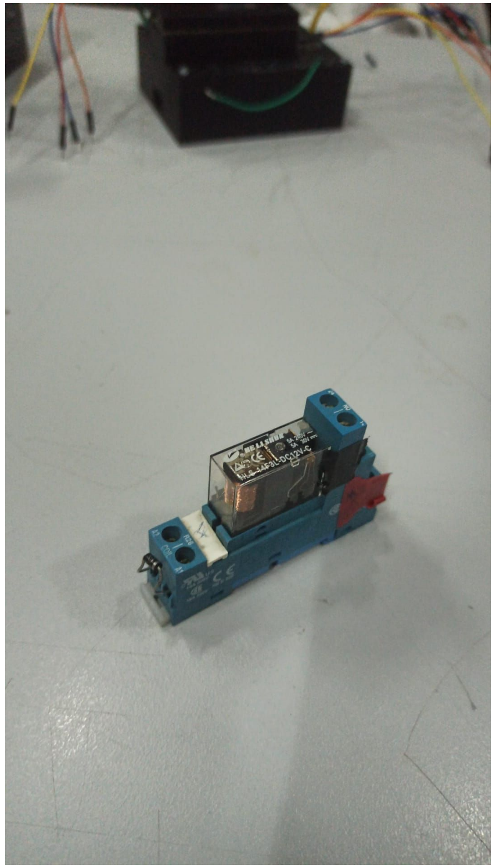
PURCHASED COMPONENTS

Name of the part:	**MPPT Charge Controller**
Part number (as in BOM):	WT50517
Quantity to be purchased:	1

PURCHASED COMPONENTS

Name of the part:	**Inverter**
Part number (as in BOM):	WT50519
Quantity to be purchased:	1

PURCHASED COMPONENTS

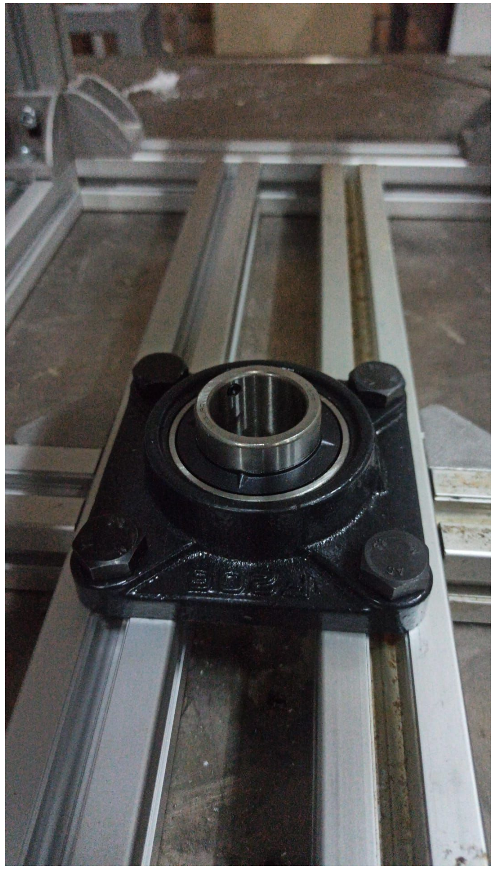


Name of the part:	**Relay**
Part number (as in BOM):	WT50521
Quantity to be purchased:	1

PURCHASED COMPONENTS

Name of the part:		**Sigma Profile**	
Part number (as in BOM):		WT40414	
Quantity to be manufactured:		1	
Material used:		Aluminium	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Cutting	Saw Machine	-
2	Drilling	Drill Press	-

PURCHASED COMPONENTS



Name of the part:	**Bearing Housing**
Part number (as in BOM):	WT30312
Quantity to be purchased:	1

PURCHASED COMPONENTS

Name of the part:	**V-Belt**
Part number (as in BOM):	WT30713
Quantity to be purchased:	1

BORROWED COMPONENTS

Name of the part:	**DC Generator**
Part number (as in BOM):	WT50518
Quantity to be borrowed:	1

BORROWED COMPONENTS

Name of the part:	**Battery**
Part number (as in BOM):	WT50520
Quantity to be borrowed:	1

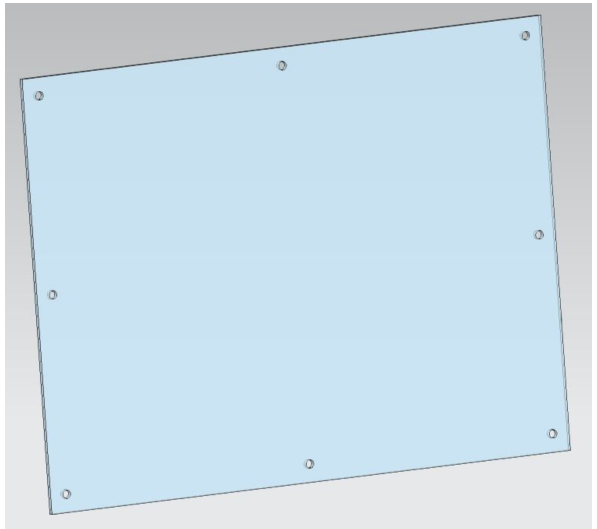
MANUFACTURED COMPONENTS
(MACHINE SHOP)

Name of the part:		Shaft	
Part number (as in BOM):		WT30110	
Quantity to be manufactured:		2	
Material used:		Iron Alloy	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Cutting	Lathe	Right Hand Turning Tool
2	Finishing	Lathe	Right Hand Turning Tool
3	Welding	TIG Weld	TIG Welding Tool

MANUFACTURED COMPONENTS
(MACHINE SHOP)

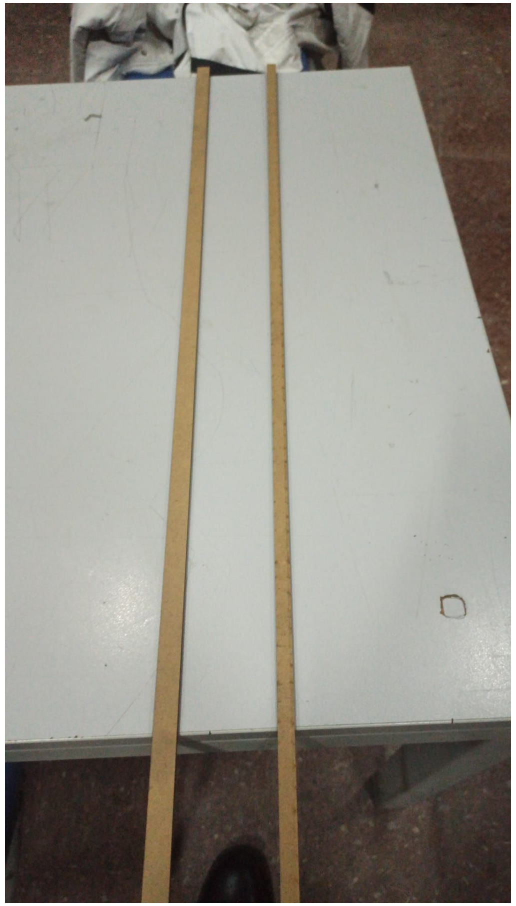
Name of the part:	**Top & Bottom Plate of Savonius**		
Part number (as in BOM):	WT20208		
Quantity to be purchased:	4		
Material used:	Plexiglass		
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-
2	Hole Drilling	Drill Press	-

MANUFACTURED COMPONENTS
(BY GROUP C2)



Name of the part:	**Side Plate of Base**		
Part number (as in BOM):	WT40315		
Quantity to be purchased:	6		
Material used:	Plexiglass		
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-
2	Hole Drilling	Drill Press	-

MANUFACTURED COMPONENTS
(MACHINE SHOP)

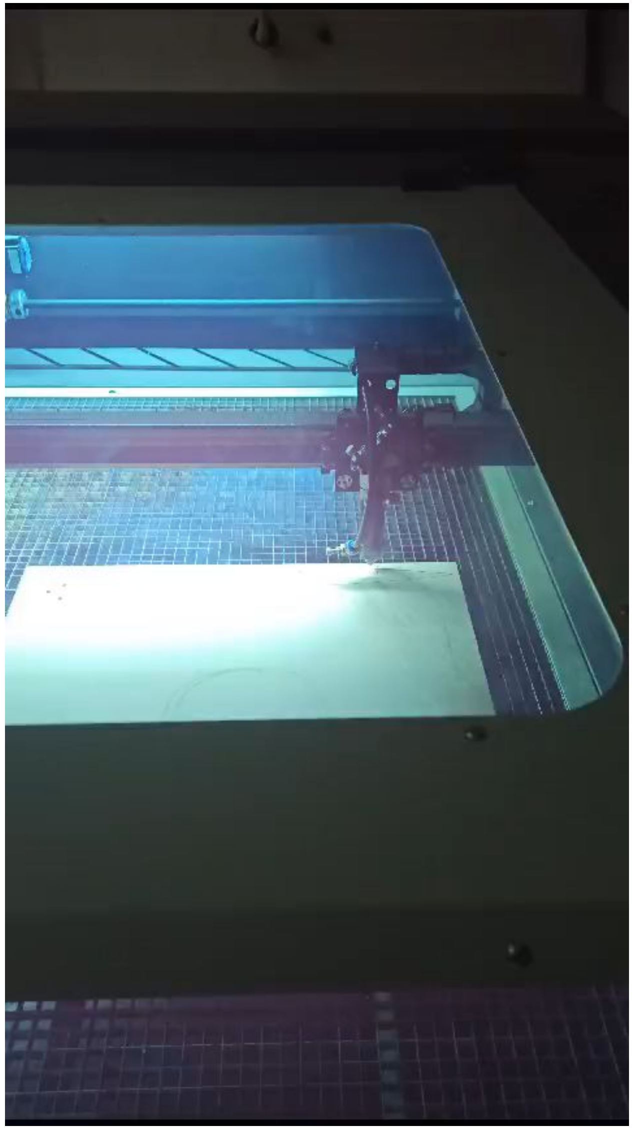


Name of the part:		**Darrius Blade Spar**	
Part number (as in BOM):		WT10101	
Quantity to be manufactured:		6	
Material used:		MDF	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-

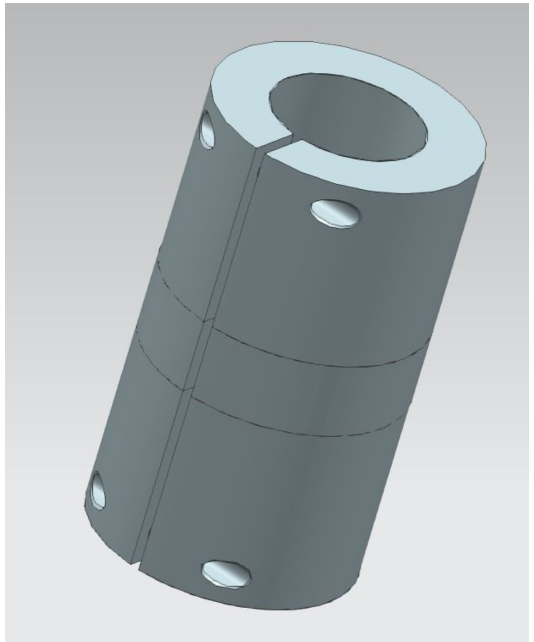
MANUFACTURED COMPONENTS
(MACHINE SHOP)

Name of the part:	Rib		
Part number (as in BOM):	WT10202		
Quantity to be manufactured:	12		
Material used:	MDF		
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-
2	Hole Drilling	Drill Press	-

MANUFACTURED COMPONENTS
(MACHINE SHOP)

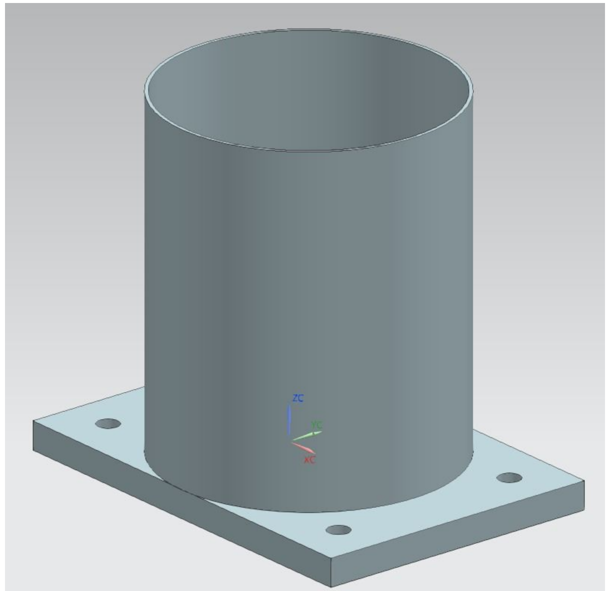


MANUFACTURED COMPONENTS
(MACHINE SHOP)



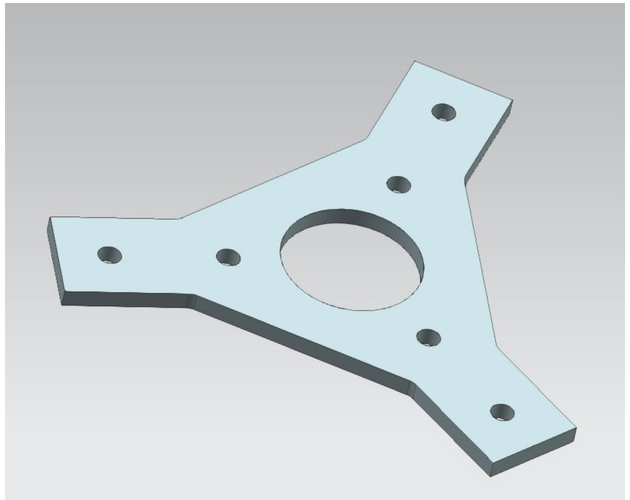
Name of the part:		Coupling	
Part number (as in BOM):		WT30311	
Quantity to be manufactured:		1	
Material used:		Cast Iron	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Drilling	Lathe	Right Hand Turning Tool
2	Threading	Lathe	Right Hand Turning Tool
3	Finishing	Lathe	Right Hand Turning Tool

MANUFACTURED COMPONENTS
(MACHINE SHOP)



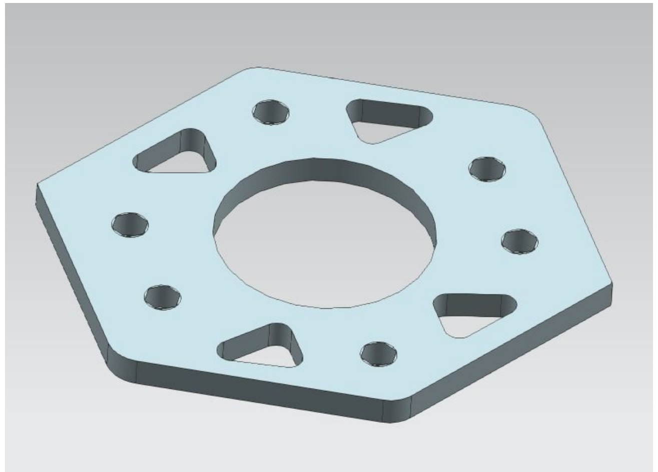
Name of the part:	**Generator Holder**		
Part number (as in BOM):	WT40416		
Quantity to be manufactured:	1		
Material used:	304 Stainless Steel Sheet Metal		
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Rolling	Rolling Machine	-
2	Welding	Weld Machine	-
3	Drilling	Drill Press	-

MANUFACTURED COMPONENTS
(OSTIM)



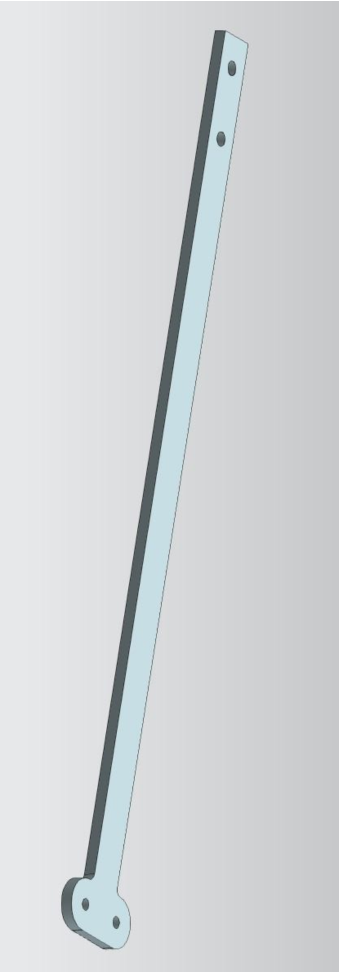
Name of the part:		Darrius Shaft Connection Component	
Part number (as in BOM):		WT10305	
Quantity to be purchased:		4	
Material used:		304 Stainless Steel Sheet Metal (4 mm)	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-
2	Hole Drilling	Drill Pres	-

MANUFACTURED COMPONENTS
(OSTIM)



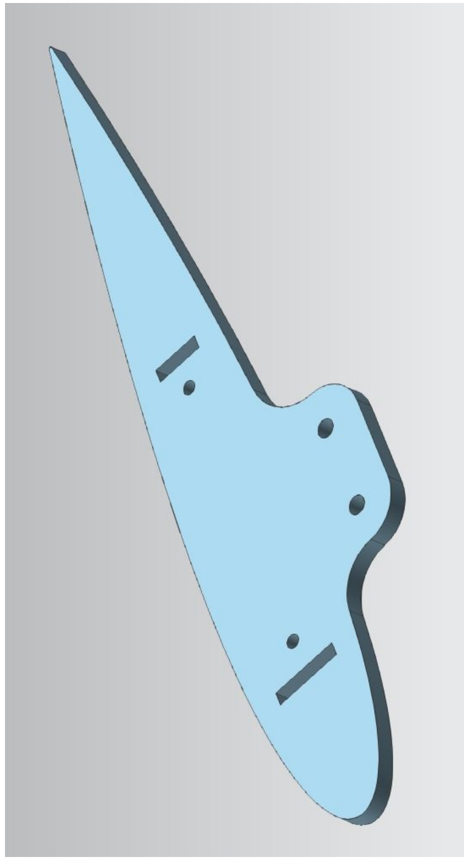
Name of the part:		**Savonius Shaft Connection Component**	
Part number (as in BOM):		WT20309	
Quantity to be purchased:		2	
Material used:		304 Stainless Steel Sheet Metal (4 mm)	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-
2	Hole Drilling	Drill Pres	-

MANUFACTURED COMPONENTS
(OSTIM)



Name of the part:		**Darrius-Shaft Connection Rod**	
Part number (as in BOM):		WT10204	
Quantity to be purchased:		6	
Material used:		304 Stainless Steel Sheet Metal (4 mm)	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Laser Cutting	Laser Cutter	-
2	Hole Drilling	Drill Pres	-

MANUFACTURED COMPONENTS
(OSTIM)



<table>
<tr><td colspan="2">Name of the part:</td><td colspan="2">Darrius Rib Connection Component</td></tr>
<tr><td colspan="2">Part number (as in BOM):</td><td colspan="2">WT1021</td></tr>
<tr><td colspan="2">Quantity to be purchased:</td><td colspan="2">6</td></tr>
<tr><td colspan="2">Material used:</td><td colspan="2">304 Stainless Steel Sheet Metal (4 mm)</td></tr>
<tr><td>Operation No:</td><td>Description of the operation</td><td>Equipment</td><td>Tooling</td></tr>
<tr><td>1</td><td>Laser Cutting</td><td>Laser Cutter</td><td>-</td></tr>
<tr><td>2</td><td>Hole Drilling</td><td>Drill Pres</td><td>-</td></tr>
</table>

MANUFACTURED COMPONENTS
(BY GROUP C2)

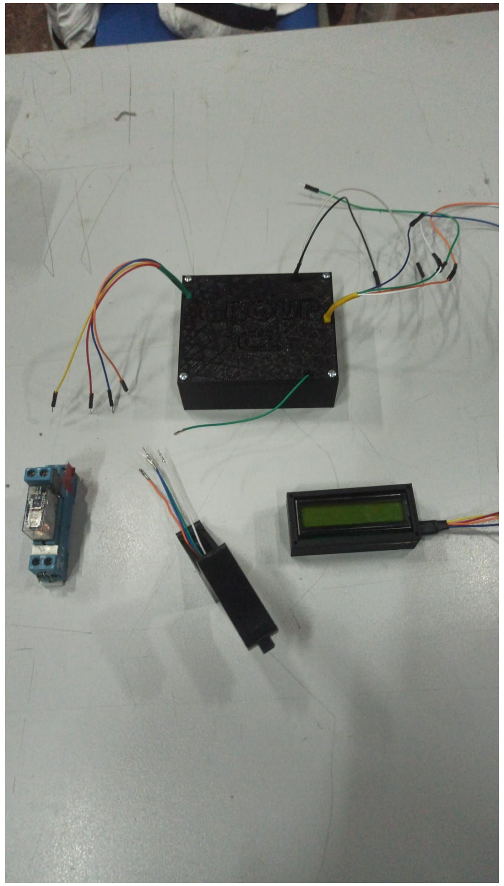
Name of the part:		**Savonius Plate**	
Part number (as in BOM):		WT20307	
Quantity to be purchased:		2	
Material used:		Aluminium composite	
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Rolling	Rolling Machine	-

MANUFACTURED COMPONENTS
(BY GROUP C2)

Name of the part:	**Darrius Blade Foam**		
Part number (as in BOM):	WT10203		
Quantity to be purchased:	30		
Material used:	Foam		
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	Foam Cutting	Hot-Wire	Metal Wire
2	Coating	Textile Coating	5D carbon fiber coating

MANUFACTURED COMPONENTS
(BY GROUP C2)

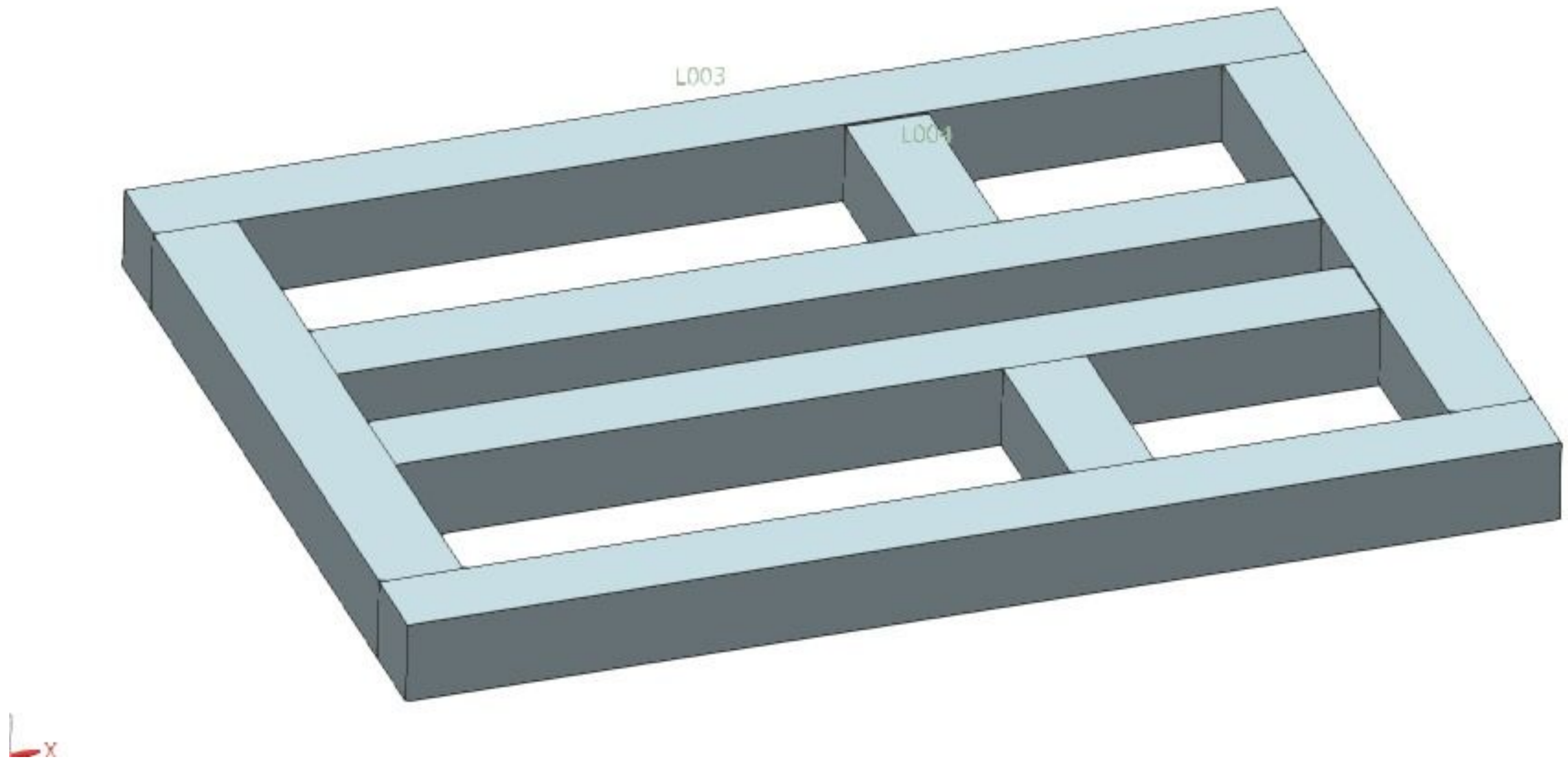
MANUFACTURED COMPONENTS
(BY GROUP C2)



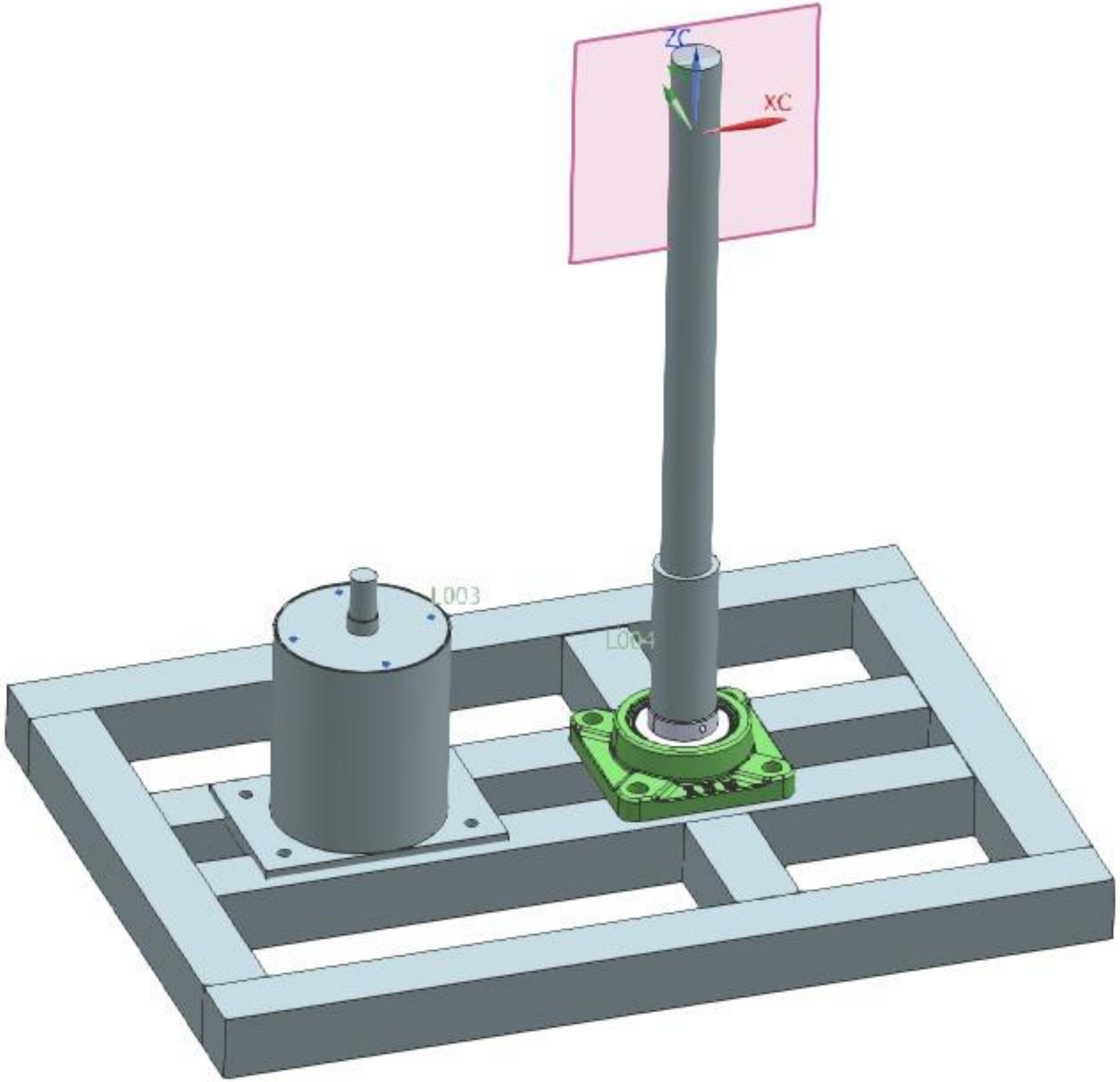
Name of the part:	Controller Enclosure		
Part number (as in BOM):	WT50521		
Quantity to be purchased:	3		
Material used:	PLA		
Operation No:	**Description of the operation**	**Equipment**	**Tooling**
1	3D Printing Process	3D Printer	-

Assembly Plan

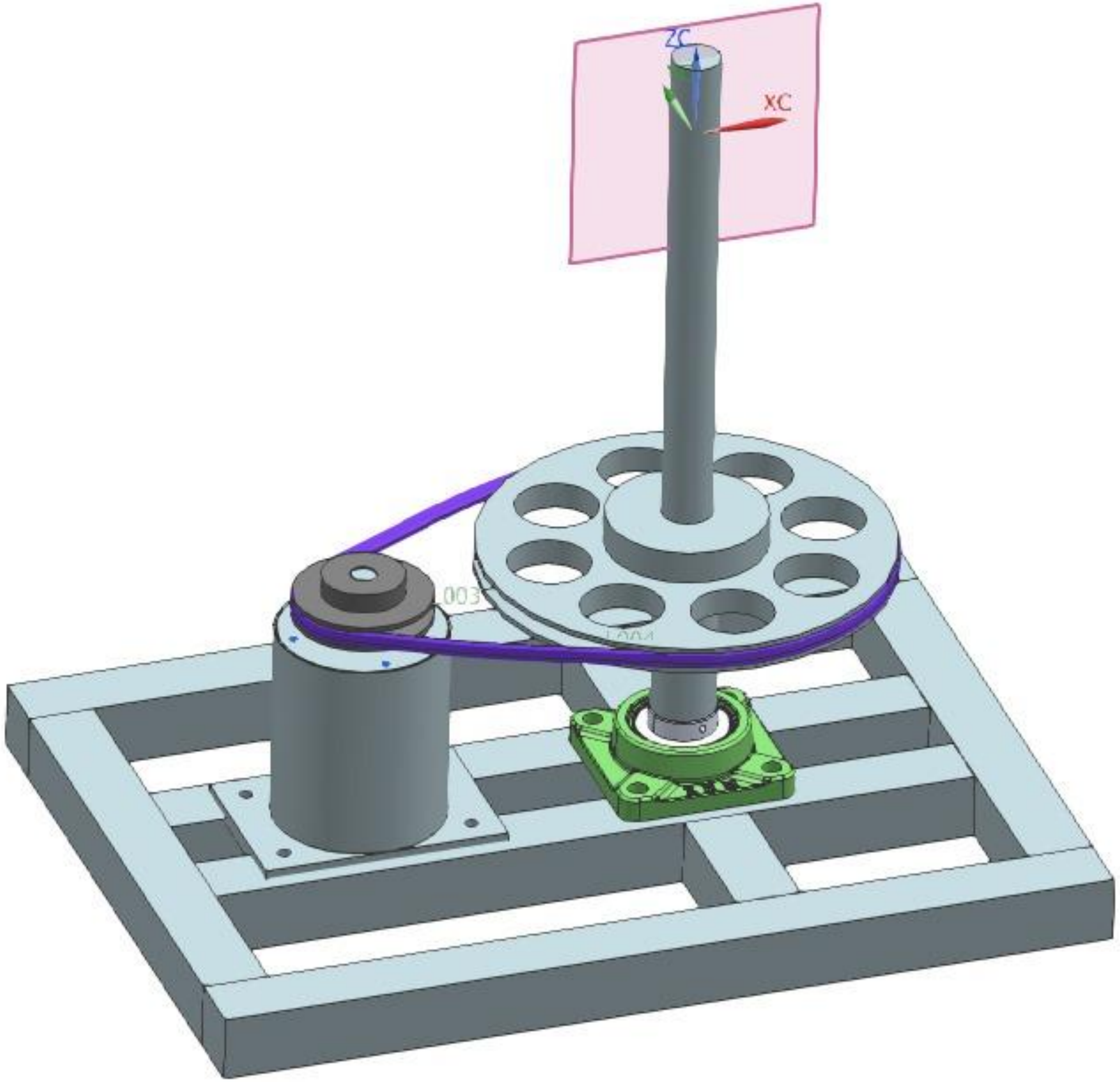
1. Base Bottom Assembly



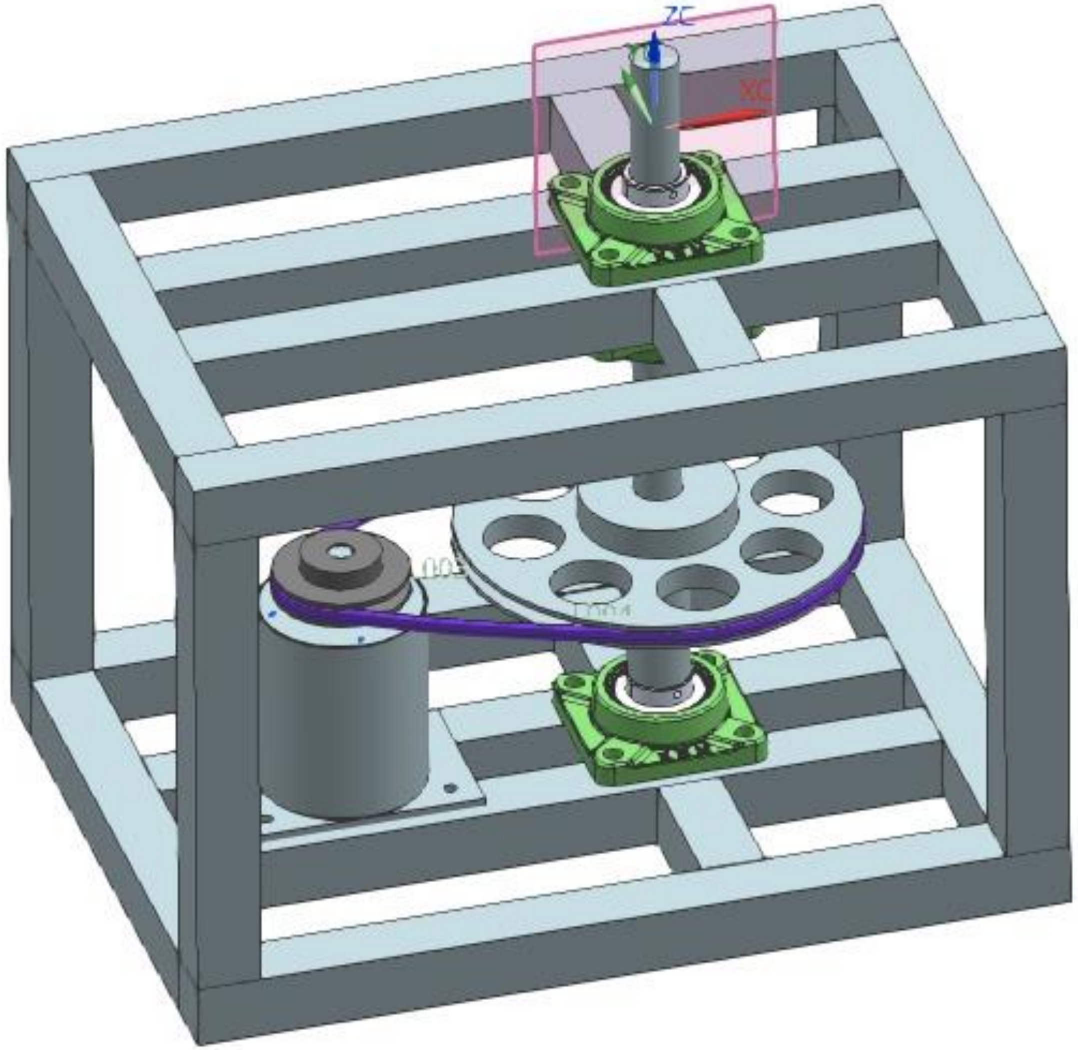
2. Base Assembly



3. V-Belt & Pulleys Assembly

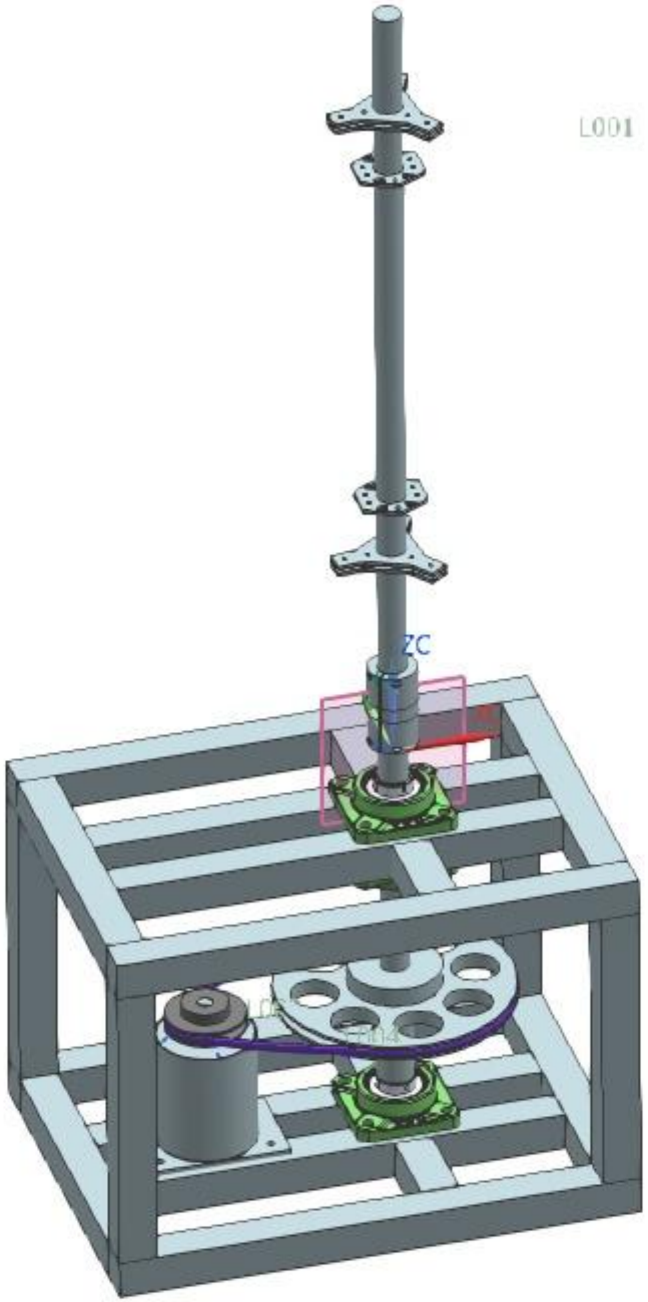


4. Base Full Assembly

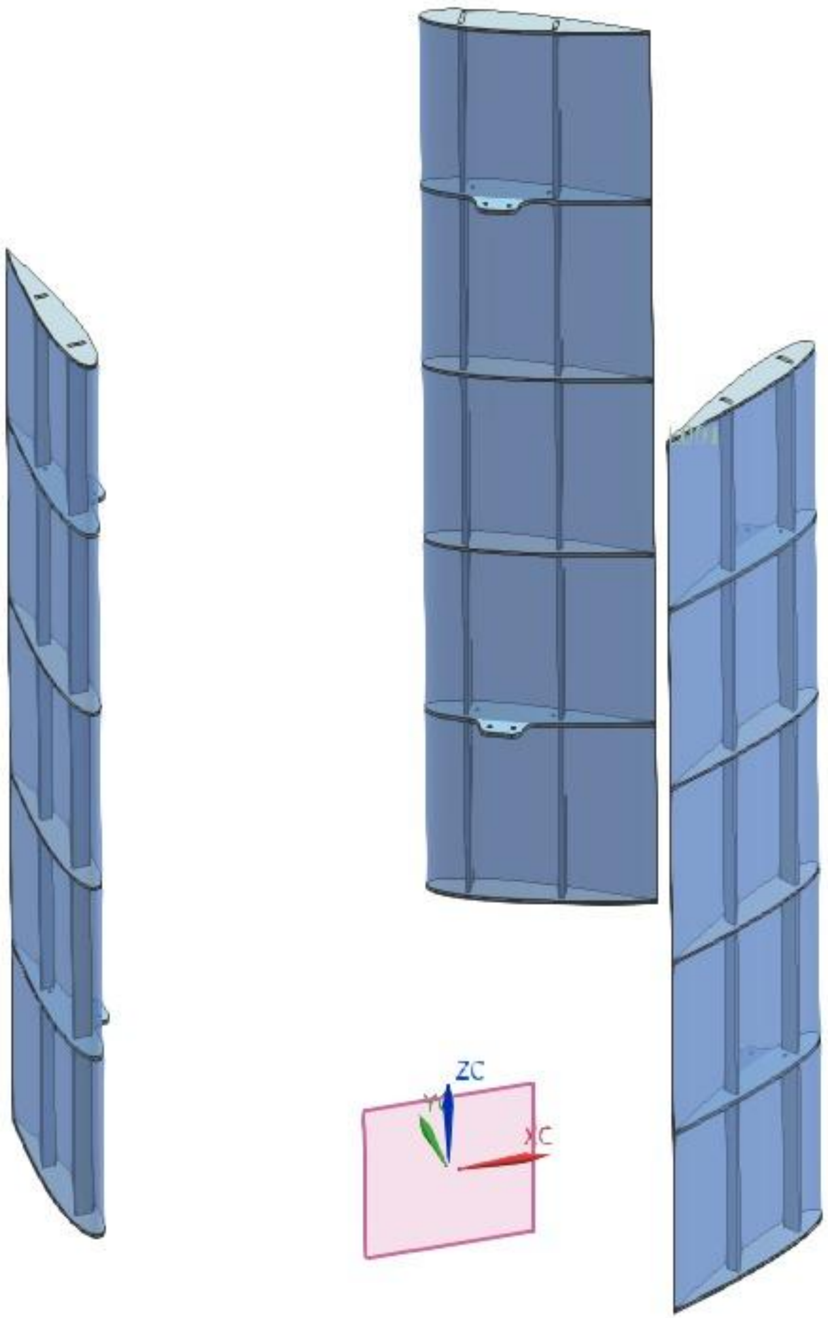


5. Upper Shaft Subassembly

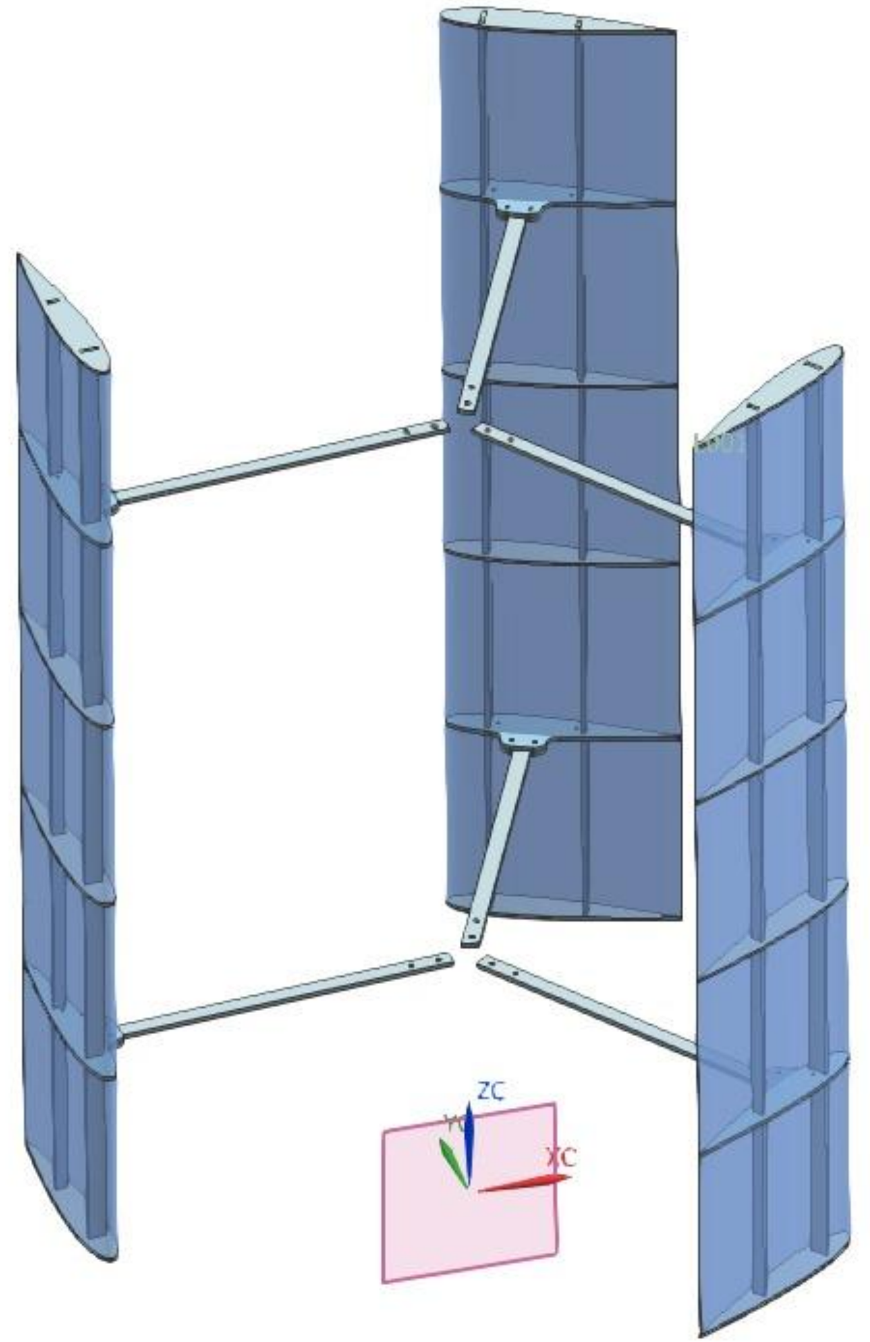
6. Shaft Connections Assembly



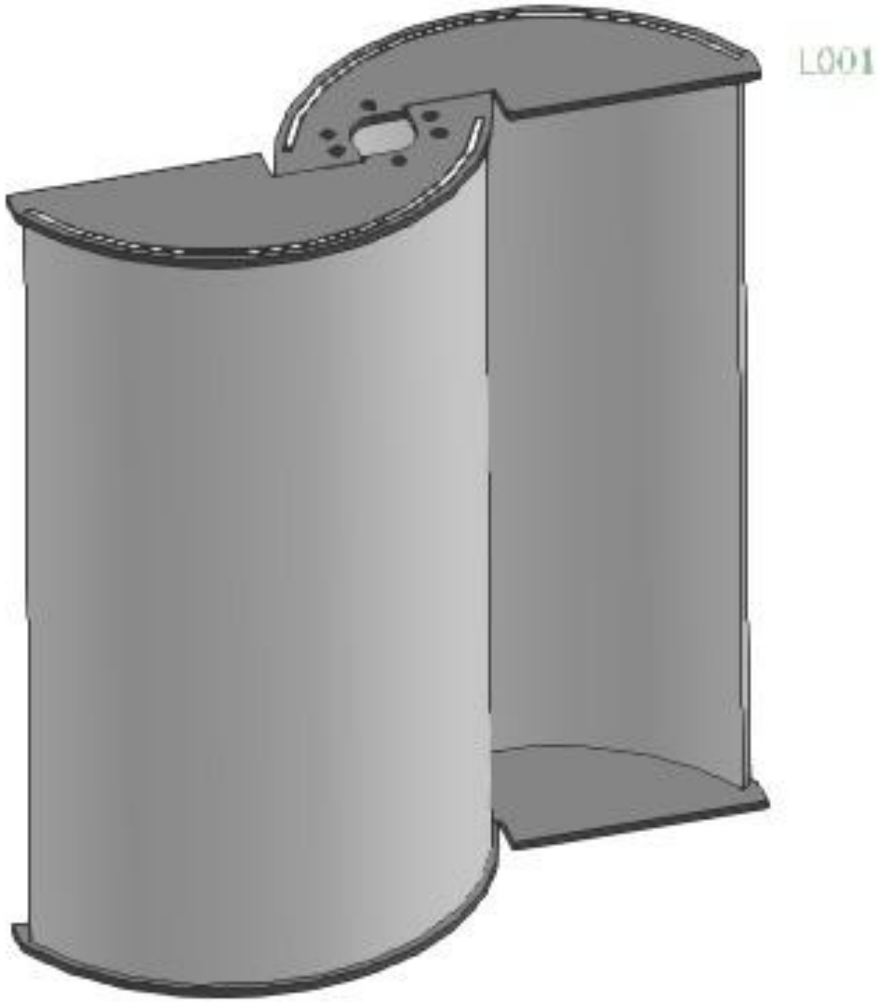
7. Darrieus Blade Assembly



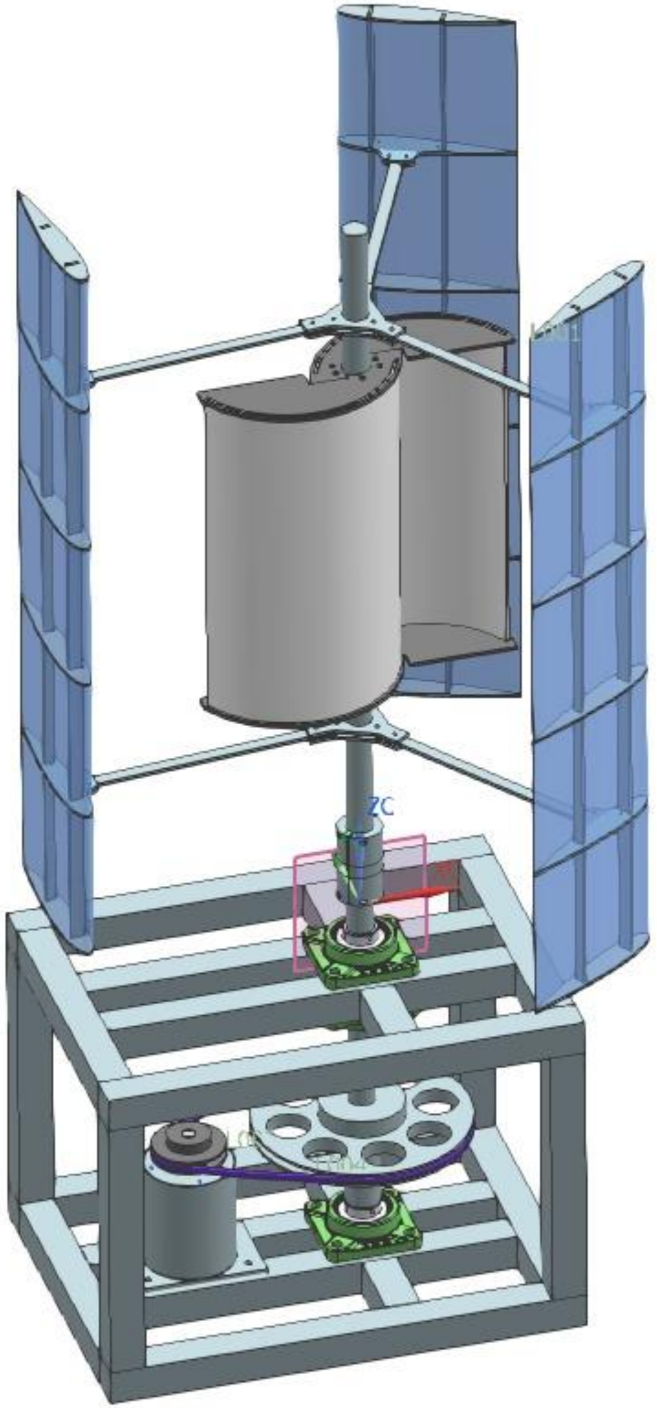
8. Darrieus Connection Assembly



9. Savonius Blade Assembly



10. Full Assembly



EXPERIENCE GAINED

- Appropriate Time Planning for The Manufacturing Process
- Team & Time Management
- Equipment Selection
- Technical Drawing with Full Assembly for Technicians

FUTURE WORK

- Full-size model
- Adjustable Savonious blade
- Sophisticated control algorithms
- The complexity of the design

Thank You For Listening